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Generalized robust loss function driven learning framework for pattern recognition

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ABSTRACT

This study introduces a novel semi-algebraic generalized robust loss function, $\mathcal{R}(x)$, integrated with the Kurdyka-Łojasiewicz (KL) property, designed for robust pattern classification. The proposed loss function uniquely combines hybrid loss mechanisms to achieve an optimal trade-off between sensitivity to minor errors and robustness against outliers while preserving computational efficiency. Within the M -estimation framework, its robustness is rigorously proven through boundedness analysis of the influence function. Extending $\mathcal{R}(x)$ to the twin support vector machine (TWSVM) framework, we develop a generalized robust TWSVM (GR-TSVM) for classification tasks. To solve the resultant non-convex, non-smooth optimization problem, an alternating direction method of multipliers (ADMM) is employed, decomposing the problem into closed-form hyperplane updates and proximal gradient steps for slack variables. Theoretically, we establish convergence guarantees and demonstrate that the objective function satisfies the KL property, ensuring stable and reliable optimization. Bridging robust statistical learning with efficient optimization, GR-TSVM addresses the limitations of quadratic loss functions and advances pattern classification by harmonizing theoretical rigor with computational tractability. Extensive comparative experiments on benchmark datasets validate the proposed method, which shows statistically significant improvements in both classification accuracy and robustness metrics compared to traditional robust learning paradigms. The code needed to reproduce all our experiments is publicly available at <https://github.com/junma1990-max/GR-TSVM-MATLAB-Demo.git>.

1. Introduction

In the era of data-driven decision-making, machine learning models are increasingly deployed in real-world applications where data quality is often compromised by noise, outliers, and measurement errors. Traditional classifiers, such as support vector machine (SVM) (Cortes & Vapnik, 1995) and twin support vector machine (TWSVM) (Jayadeva et al., 2007), rely on convex loss functions (e.g., quadratic or hinge loss) that are inherently sensitive to such perturbations. While these methods achieve remarkable performance under idealized conditions, their fragility in noisy environments poses significant risks to safety-critical systems (Huber, 1964; Maronna et al., 2019). This limitation has spurred the development of robust learning paradigms that prioritize stability and reliability over mere predictive accuracy.

Robust machine learning, at its core, seeks to develop models resilient to data imperfections outliers, noise, and distributional shifts by strategically redesigning loss functions and regularization mechanisms (Koenker & Bassett, 1978; Liu et al., 2007; Rousseeuw & Leroy, 1987).

This field has evolved through iterative breakthroughs and persistent challenges, shaped by the tension between theoretical robustness, computational feasibility, and adaptability to increasingly complex data environments. The genesis of robust learning lies statistics, where Huber's seminal M -estimators redefined error penalization by replacing quadratic loss with a hybrid function quadratic for small residuals and linear for large deviations (Huber, 1964). This bifurcated design, formalized as

$$L_{\delta}(r) = \begin{cases} \frac{1}{2}r^2 & \text{if } |r| \leq \delta, \\ \delta|r| - \frac{1}{2}\delta^2 & \text{otherwise,} \end{cases}$$

where r is the residual and δ a tunable threshold, achieved outlier resistance while maintaining convexity for tractable optimization. Parallel innovations like Rousseeuw's trimmed losses took a more aggressive approach, discarding a fixed percentage of extreme samples during estimation (Rousseeuw & Leroy, 1987). These methods excelled in low-dimensional linear regression, where closed-form solutions and asymptotic normality could be guaranteed under mild contamination

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assumptions (Maronna et al., 2019). However, the efficacy of Huber and trimmed losses hinged on correctly setting hyperparameters (δ , trimming proportion) a task requiring prior knowledge of outlier severity, which is often unavailable in practice. In heterogeneous noise regimes, static thresholds failed to adapt, either over-penalizing valid samples or under-suppressing outliers. Non-convex alternatives like Tukeys bi-weight (Lu et al., 2022) or Cauchy losses (Li et al., 2018), which saturate completely for large residuals, offered stronger outlier immunity but introduced non-convex optimization landscapes riddled with suboptimal local minima. For instance, Tukeys loss, defined as

$$L(r) = \begin{cases} \frac{c^2}{6} \left[1 - \left(1 - \left(\frac{r}{c} \right)^2 \right)^3 \right] & \text{if } |r| \leq c, \\ \frac{c^2}{6} & \text{otherwise,} \end{cases}$$

where c is a tuning constant, provided complete outlier rejection beyond $|r| > c$. Yet, its non-convexity necessitated careful initialization and iterative reweighting schemes, complicating large-scale deployment. Classical methods assumed $n \gg d$ (sample size $\gg$ features), where outlier effects could be averaged out. In high dimensions ($d \approx n$ or $d \gg n$), however, M-estimators suffered from inflated variance, causing even mild contamination to catastrophically distort parameter estimates a phenomenon formalized in their analysis of robust regression breakdown points under dimensionality scaling. Correntropy-induced loss (C-Loss) (Liu et al., 2007) leveraged kernel methods to map residuals into Reproducing Kernel Hilbert Spaces (RKHS), dynamically weighting samples based on similarity to the core data distribution. For Gaussian kernels, C-Loss computes weights as $w_i = \exp(-r_i^2/2\sigma^2)$, where σ controls the bandwidth. Similarly, quantile regression employed asymmetric absolute loss $L_\tau(r) = r(\tau - \mathbb{I}_{r < 0})$ to estimate conditional quantiles, proving invaluable in heteroscedastic scenarios like financial risk modeling (Koenker & Bassett, 1978). However, these methods introduced new dilemmas. Kernel-based C-Loss requires computing an $n \times n$ Gram matrix, incurring $O(n^2)$ time and memory complexity. Even with approximations like Nyström sampling, datasets exceeding 10^5 samples remained intractable (Zhang et al., 2017). Quantile regression, while interpretable, lacked guarantees in non-i.i.d. settings. Adversarial contamination or feature-dependent noise often violated its assumptions, leading to inconsistent estimates (Koenker & Bassett, 1978). Both approaches treated residuals as scalar quantities, ignoring multivariate outlier structures. In genomics or computer vision, where anomalies manifest as coordinated perturbations across feature groups, univariate robust losses proved inadequate (Zhang et al., 2017). This limitation has spurred the development of robust learning paradigms that prioritize stability and reliability over mere predictive accuracy.

SVM (Cortes & Vapnik, 1995) is a cornerstone of statistical learning theory, are renowned for their structural risk minimization principle and robustness in high-dimensional, small-sample regimes. To further enhance their performance and robustness, research efforts have led to the development of modified SVM formulations. Wu and Liu (2007) proposed a robust truncated hinge loss SVM, which truncates the hinge loss to mitigate outlier effects. Wang et al. (2008) proposed a primal-space smooth ramp loss SVM, facilitating efficient optimization and graceful handling of misclassifications. Huang et al. (2013) proposed a pinball loss SVM, offering flexible error balancing across classes; and Shen et al. (2017) truncated pinball loss SVM, which synergizes truncation and pinball loss advantages for improved robustness and generalization. In large-scale pattern classification task, Wang et al. (2024a) proposed a sparse and robust support vector machine with capped squared loss. Wang and Shao (2023, 2024) proposed fast generalized ramp loss support vector machine and fast truncated Huber loss SVM for large scale classification. A SVM classifier via $L_{0/1}$ soft-margin loss proposed for classification task by Wang et al. (2021). These advancements underscore the continuous refinement of SVMs to address real-world classification challenges with enhanced theoretical rigor and practical efficacy. Despite their remarkable classification performance, SVMs face notable

computational burdens. To mitigate the impact of penalty levels for correctly classified samples within the margin region on the generalization performance of SVM classifiers, Li and Zhang (2024) introduced a novel Slide loss function, which dynamically adjusts penalties based on sample proximity to the decision boundary. Subsequently, Li and Yang (2024) introduced a bounded exponential quantile loss for SVM classification. Their experimental results on multiple real-world datasets demonstrate the superior robustness and classification efficacy of the proposed Slide loss compared to conventional hinge loss formulations. Motivated by quantile regression in statistical theory and the bounded linex loss function, Zhang and Yang (2024) proposed a novel bounded quantile loss to enhance the robustness of SVM. This loss function inherits key properties such as asymmetry and non-convexity that enable SVM models to effectively suppress noise and outliers while maintaining discriminative power. To address the high computational complexity of traditional support vector machines (SVMs) in large-scale classification tasks, Wang et al. (2024c) proposed a novel truncated squared loss function, resulting in a more efficient SVM variant. However, the resulting model presents significant optimization challenges due to its inherent nonconvexity and nonsmoothness, requiring specialized algorithmic approaches for effective training. Zhao et al. (2025) propose a novel Quadratic-Type Squared Error Loss Function (QTSELF), which is continuous, smooth, and differentiable, and employ it to develop a more robust One-Class Support Vector Machine (Q-OCSVM). Specifically, SVM training involves solving a quadratic programming (QP) problem with a complexity of $O(n^3)$, leading to prohibitive time and resource consumption for large-scale datasets. Traditional optimization algorithms suffer from slow convergence on ultra-large datasets and lack efficient parallelization. In high-dimensional sparse data scenarios, the quadratic complexity of kernel matrix operations and inefficient exploitation of sparsity further constrain scalability. Collectively, these challenges hinder SVMs' practical deployment in big data and real-time applications.

To address SVM's computational bottleneck, Jayadeva et al. (2007) proposed the twin SVM (TWSVM), which decomposes the single large-scale optimization into two class-specific quadratic programming problems (QPPs) of complexity $O((n/2)^3)$ each. Unlike SVM's single maximal-margin hyperplane, TWSVM constructs asymmetric, non-parallel hyperplanes by maximizing each class's proximity to its own support vectors while minimizing overlap with the opposing class, achieving a theoretical 4-fold speedup over standard SVM. In recent years, some robust learning methods have been proposed for pattern classification based on TWSVM. In order to improve the performance and robustness of TWSVM, the researchers developed improved TWSVM formulations. Shao et al. (2015) proposed a weighted linear loss TWSVM (WLTSVM) for large-scale classification. Xu et al. (2017) proposed a novel TWSVM with pinball loss for robust classification task. Tanveer et al. (2019) proposed a novel sparse TWSVM with pinball loss for robust classification task. Yan et al. (2017) proposed a new L_1 -GEPSVM classifier based on an effective iterative algorithm for classification. Li et al. (2015) proposed a robust L_1 -norm non-parallel proximal SVM (L_1 -NPSVM) for pattern classification. A new robust capped L_1 -norm TWSVM (CTWSVM) proposed by Wang et al. (2019). Capped L_1 -norm distance metric-based fast robust twin bounded support vector machine for pattern classification proposed by Ma et al. (2020). Ma et al. (2021) proposed a novel adaptive robust learning framework (ARTSVM) for TWSVM classification. Yuan and Yang (2021) proposed a capped $L_{2,p}$ -norm metric based robust least squares TWSVM for robust pattern classification. To address the challenges of outlier prone and high-dimensional data which often lead to suboptimal performance and slow convergence, Akhtar et al. (2024c) proposed a novel robust, bounded, sparse, and smooth (RoBoSS) loss function for supervised learning. Akhtar et al. (2024c) provided a theoretical analysis of the RoBoSS loss, including its classification-calibrated property and generalization guarantees. These results offer key insights into its robustness in classification tasks and its ability to generalize effectively to unseen data. Subsequently, Akhtar et al. (2024a) introduced the wave loss, an asymmetric loss function for supervised learn-

ing that demonstrates robustness to outliers, noise tolerance, boundedness, and smoothness. Theoretically, we prove that this loss function satisfies the key property of classification calibration. To address the sensitivity of TSVM to outliers and noise a limitation inherent to hinge loss, Akhtar et al. (2024b) proposed the Guardian loss (G-loss), a novel asymmetric loss function characterized by boundedness, smoothness, and inherent robustness. At the same time, via integrating G-loss into TWSVM, Akhtar et al. (2024b) developed GL-TSVM, a classifier that combines the discriminative power of TWSVM with enhanced robustness and smooth optimization properties. Recently, many scholars have proposed novel learning methods based on the algorithm of dual support vector machines, such as granular-ball fuzzy twin support vector machine (GBFTSVM) (Lang et al., 2025), robust capped L1-norm kernel-free quadratic surface twin support vector machine (CLIQTSVM) (Si et al., 2025), Granular Ball K-Class Twin Support Vector Classifier (GB-TWKSVC) (Ganaie et al., 2025), least squares twin bounded support vector machine with a generalized correntropy-induced metric (CLSTB-SVM) (Yuan et al., 2025), double well potential twin support vector regression (DWPTSVM) (Ma et al., 2025), robust granular ball least square twin support vector machine (GBLSTVM) (Tanveer et al., 2025).

In this paper, we propose a novel semi-algebraic generalized robust loss function with Kurdyka-Łojasiewicz property for pattern classification, denoted as $\mathcal{R}(x)$, which is a novel framework that combines hybrid loss functions with theoretically grounded optimization methods. The core innovation of this loss function lies in its hybrid loss function, which achieves an optimal balance between sensitivity to small errors and robustness against large deviations, while maintaining computational efficiency. Within the framework of M -estimation, the robustness of the loss function $\mathcal{R}(x)$ is strictly proven by evaluating the boundedness of its influence function. Subsequently, we extend the $\mathcal{R}(x)$ to TWSVM learning framework, developing a generalized robust TWSVM (GR-TSVM) framework for pattern classification tasks. To optimize the resulting non-convex and non-smooth objective function, we employ an Alternating Direction Method of Multipliers (ADMM) framework (Boyd et al., 2011; Guan et al., 2018; Wang et al., 2024b). This approach decomposes the optimization problem into two manageable components closed-form updates for hyperplane weights and proximal gradient steps for slack variables (Parikh & Boyd, 2014; Wang et al., 2022). Crucially, we provide a rigorous theoretical analysis of the proposed model, establishing convergence guarantees and proving that the objective function satisfies the Kurdyka-Łojasiewicz (KL) property (Attouch et al., 2010, 2013; Bolte et al., 2014). This ensures the stability and reliability of the optimization process. By bridging the gap between robust statistical learning and efficient optimization, this study offers a principled and practical solution for real-world applications, particularly in scenarios where data integrity cannot be guaranteed. The proposed framework not only addresses the limitations of quadratic loss functions but also advances the state-of-the-art in pattern classification by combining theoretical rigor with computational tractability. This motivation highlights the necessity of GR-TSVM in addressing the limitations of existing methods while emphasizing its theoretical and practical contributions to robust machine learning. The proposed method is further validated through extensive comparative experiments on benchmark datasets, demonstrating statistically significant improvements in classification accuracy and robustness metrics compared to conventional robust learning paradigms.

The main contributions of our paper are summarized as follows:

- (1) A novel generalized robust loss function, denoted as $\mathcal{R}(x)$, is introduced. The robustness, boundedness, and nonconvexity of $\mathcal{R}(x)$ are analyzed and demonstrated in detail. This formulation incorporates logarithmic error suppression for large deviations, a perception-inspired saturating fractional term, and L_1 sparsity induction to filter irrelevant features.
- (2) Through extension of $\mathcal{R}(x)$ to the TWSVM learning framework, a generalized robust TWSVM (GR-TSVM) is developed for classification problems.

- (3) To solve the generalized robust twin support vector machine (GR-TSVM) optimization problem, an alternating direction method of multipliers (ADMM) is employed. The optimization problem is decomposed into closed-form hyperplane updates and proximal gradient steps for slack variables.

- (4) Theoretically, convergence guarantees are established for the proposed method, and the objective function is demonstrated to satisfy the Kurdyka-Łojasiewicz property, which ensures stable and reliable optimization.

- (5) The proposed method GR-TSVM is validated through extensive comparative experiments on benchmark datasets. Statistically significant improvements in both classification accuracy and robustness metrics are demonstrated compared to traditional robust learning paradigms.

The remainder of this paper is structured as follows: In Section 1, we reviews related work on Cauchy loss function and TWSVM. In Section 3, we give the proposed work of this paper, which mainly consist of robust loss function $\mathcal{R}(x)$, GR-TSVM, convergence analysis and complexity analysis. In Section 4, we conduct a series of numerical experiments based on benchmark datasets and present a comprehensive analysis. In Section 5, we give the conclusions of this paper and directions for future research.

2. Related work

2.1. Loss functions

2.1.1. Cauchy loss function

Traditional machine learning models employing convex loss functions exhibit heightened sensitivity to data imperfections, where quadratic loss particularly amplifies outlier effects, inducing parameter bias and compromised generalization. Such vulnerability restricts their deployment in safety-critical applications requiring high reliability, including medical diagnostics and autonomous systems. Although robust alternatives like M -estimators and correntropy loss have emerged to mitigate these limitations, critical challenges persist: (1) inherent tensions between non-convex formulations and computational efficiency, (2) scalability constraints with high-dimensional data, and (3) insufficient theoretical guarantees regarding convergence and generalization bounds. In recent years, the Cauchy loss function from the heavy-tailed Cauchy distribution, which enhances the robustness of outliers by logarithmically attenuating the influence of large residuals, has received extensive attention. The Cauchy loss function is formally defined as

$$L(\delta) = \ln \left(1 + \frac{\delta^2}{\theta^2} \right), \quad (1)$$

where δ represents the prediction error and $\theta > 0$ is a scale parameter governing the loss's sensitivity to residuals. By leveraging the logarithmic suppression of large errors, this function inherently limits outlier influence compared to conventional quadratic or hinge losses, making it particularly advantageous in applications like medical imaging or sensor fusion where data corruption is prevalent. Its adaptive thresholding via the scale parameter $\theta > 0$ eliminates manual tuning, while smooth gradients stabilize optimization compared to non-convex alternatives. However, its non-convexity introduces multiple local minima, necessitating careful initialization or adaptive learning strategies to avoid suboptimal convergence. Performance is sensitive to $\theta > 0$, with suboptimal scaling risking underfitting or insufficient outlier rejection, and flattened gradients for large errors can prolong training. Theoretical gaps in convergence guarantees and generalization bounds, alongside suboptimal performance in low-noise settings, further limit its adoption in safety-critical domains. This reflects a fundamental trade-off, while the Cauchy loss excels in outlier resilience, it inherits challenges inherent to non-convex optimization and lacks rigorous theoretical underpinnings for broader applicability.

2.1.2. RoBoSS loss function

Akhtar et al. (2024a) proposed a novel robust, bounded, sparse, and smooth loss function for supervised learning. To enhance the robustness, sparsity, and smoothness compared to existing losses, they introduced the RoBoSS loss function, defined as

$$\mathcal{L}_{\text{RoBoSS}}(u) = \begin{cases} \lambda\{1 - (au + 1)\exp(-au)\}, & u > 0, \\ 0, & u \leq 0, \end{cases} \quad (2)$$

where $a, \lambda > 0$ represent the shape and bound parameters, respectively. The RoBoSS loss function is designed to address multiple critical challenges in supervised learning simultaneously:

- **Robustness:** It ensures stable learning by mitigating the influence of outliers.
- **Boundedness:** By restricting the impact of extreme values, it prevents unbounded loss growth, enhancing numerical stability.
- **Sparsity:** It selectively focuses on misclassified or boundary-proximate samples, promoting model parsimony and efficient feature utilization.
- **Smoothness:** Its smooth formulation enables gradual optimization, avoiding abrupt parameter updates and ensuring stable convergence during training.

2.1.3. Wave loss function

The wave loss $\mathcal{L}_{\text{wave}}(u)$ (Akhtar et al., 2024a) offers three key advantages: robustness to outliers, insensitivity to noise, and inherent smoothness. The function is formally defined as

$$\mathcal{L}_{\text{wave}}(u) = \frac{1}{\lambda} \left(1 - \frac{1}{1 + \lambda u^2 e^{au}} \right), \quad \forall u \in \mathbb{R}, \quad (3)$$

where $a \in \mathbb{R}$ is shape parameter and $\lambda \in \mathbb{R}^+$ is bounding parameter. The wave loss function $\mathcal{L}_{\text{wave}}(u)$, showcases the following inherent properties:

- The $\mathcal{L}_{\text{wave}}(u)$ exhibits inherent robustness to outliers by bounding the maximum loss to $1/\lambda$, thereby preventing extreme errors from dominating the optimization process. Simultaneously, its non-zero penalty for samples with $u \leq 0$ enhances noise insensitivity, as it maintains informative gradients even for potentially mislabeled or noisy data points.
- The $\mathcal{L}_{\text{wave}}(u)$ is non-convex, smooth, and bounded.
- The $\mathcal{L}_{\text{wave}}(u)$ is infinitely differentiable and hence continuous for all $u \in \mathbb{R}$.
- For given value of λ , when $a \rightarrow +\infty$, the $\mathcal{L}_{\text{wave}}(u)$ converges pointwise to the $0 - 1/\lambda$ loss which is defined as

$$\mathcal{L}_{0-1}(u) = \begin{cases} \frac{1}{\lambda}, & u > 0, \\ 0, & u \leq 0. \end{cases}$$

Furthermore, when $\lambda = 1$ it converges to $0 - 1$ loss.

- The shape parameter (a) determines the loss function's shape, and bounding parameter (λ) sets thresholds for loss values.

2.1.4. Guardian loss function

To enhance the robustness of traditional learning algorithms by mitigating the impact of outliers and noise, Akhtar et al. (2024a) propose a novel asymmetric loss function, termed the Guardian Loss (G-Loss). The G-Loss is meticulously formulated to possess three key properties: asymmetry, boundedness, and smoothness. Its mathematical representation is given by

$$\mathcal{L}_G(r) = 1 - \frac{1}{1 + r\{\exp(ar) - 1\}}, \quad \forall r \in \mathbb{R}, \quad (4)$$

where $r = 1 - yf(x)$ and $a > 0$ governs the asymmetry of the G-loss.

The $\mathcal{L}_G(r)$ is asymmetric, bounded, and smooth function:

- *Guides algorithms* during training through its directional properties.

- *Assigns distinct penalties* to samples based on their position relative to the proximal hyperplane.
- *Bounds maximum loss* for deviant points, curtailing outlier/noise influence.
- *Enables gradient-based optimization* through continuous differentiability.

2.2. Twin support vector machine

Although SVM is an efficient pattern classification tool. However, SVM needs to solve a large scale QP problem, which greatly affects its generalization capability. To overcome this difficulty effectively, Jayadeva et al. (2007) proposed a twin support vector machine (TWSVM), whose goal is to generate two nonparallel planes by solving a pair of smaller QPPs such that each plane is closer to one class and further from the other.

Considering n -dimensional Euclidean space $\mathbb{R}^n$ the binary classification problem, we set the training set is $\mathcal{T}_l = \{(x_1, y_1), \dots, (x_l, y_l)\} \in (\mathbb{R}^n, \mathcal{Y})^l$, where $x_i \in \mathbb{R}^n$, $y_i \in \mathcal{Y} = \{1, -1\}$, $i = 1, \dots, l$. $\mathcal{T}_l$ contains m_1 positive class and m_2 negative class, where $l = m_1 + m_2$. Let $\mathbf{A} \in \mathbb{R}^{m_1 \times n}$ represents all positive class samples, where each row $A_i \in \mathbb{R}^n$, and $\mathbf{B} \in \mathbb{R}^{m_2 \times n}$ represents all negative class samples, where each row $B_i \in \mathbb{R}^n$. TWSVM identifies two non-parallel hyper planes in the feature space

$$f_1(x) = \omega_1^T x + b_1 = 0, \quad (5)$$

$$f_2(x) = \omega_2^T x + b_2 = 0, \quad (6)$$

where $\omega_1 \in \mathbb{R}^n, \omega_2 \in \mathbb{R}^n, b_1 \in \mathbb{R}, b_2 \in \mathbb{R}$.

The TWSVM classifier is obtained by solving the following pair of QPPs

$$\min_{\omega_1, b_1} \frac{1}{2} \|\mathbf{A}\omega_1 + \mathbf{e}_1 b_1\|_2^2 + C_1 \mathbf{e}_1^T \xi_1, \quad (7)$$

$$s.t. \quad -(\mathbf{B}\omega_1 + \mathbf{e}_2 b_1) + \xi_1 \geq \mathbf{e}_2, \quad \xi_1 \geq 0. \quad (8)$$

$$\min_{\omega_2, b_2} \frac{1}{2} \|\mathbf{B}\omega_2 + \mathbf{e}_2 b_2\|_2^2 + C_2 \mathbf{e}_2^T \xi_2, \quad (9)$$

$$s.t. \quad (\mathbf{A}\omega_2 + \mathbf{e}_1 b_2) + \xi_2 \geq \mathbf{e}_1, \quad \xi_2 \geq 0. \quad (10)$$

where $C_1 \geq 0, C_2 \geq 0$ are regularization parameters, $\mathbf{e}_1$ and $\mathbf{e}_2$ are column vectors of ones of appropriate dimensions, ξ_1, ξ_2 are the slack vectors.

Generally, by introducing the lagrangian multipliers α and β , and let $\mathbf{H} = [\mathbf{A}, \mathbf{e}_1]$ and $\mathbf{G} = [\mathbf{B}, \mathbf{e}_2]$, we can get the dual problem of TWSVM via dual theory:

$$\min_{\alpha} \frac{1}{2} \alpha^T \mathbf{G}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{G}^T \alpha - \mathbf{e}_2^T \alpha, \quad (11)$$

$$s.t. \quad 0 \leq \alpha \leq C_1 \mathbf{e}_2. \quad (12)$$

$$\min_{\beta} \frac{1}{2} \beta^T \mathbf{H}(\mathbf{G}^T \mathbf{G})^{-1} \mathbf{H}^T \beta - \mathbf{e}_1^T \beta, \quad (13)$$

$$s.t. \quad 0 \leq \beta \leq C_2 \mathbf{e}_1. \quad (14)$$

Furthermore, we can get nonlinear TWSVM by introducing kernel method, and can be decided whether the sample data x belongs to positive class or negative class according to the shortest distance between the sample data x and two non-parallel planes. The decision function is

$$f(x) = \arg \min_{k=1,2} \frac{|x\omega_k + b_k|}{\|\omega_k\|}. \quad (15)$$

3. Proposed work

3.1. Motivation

The motivation of this paper stems from three critical limitations in existing pattern classification methodologies: (1) Conventional quadratic loss functions exhibit inherent sensitivity to minor errors while lacking robustness against outliers, creating a suboptimal trade-off between precision and noise resilience. (2) Traditional twin support

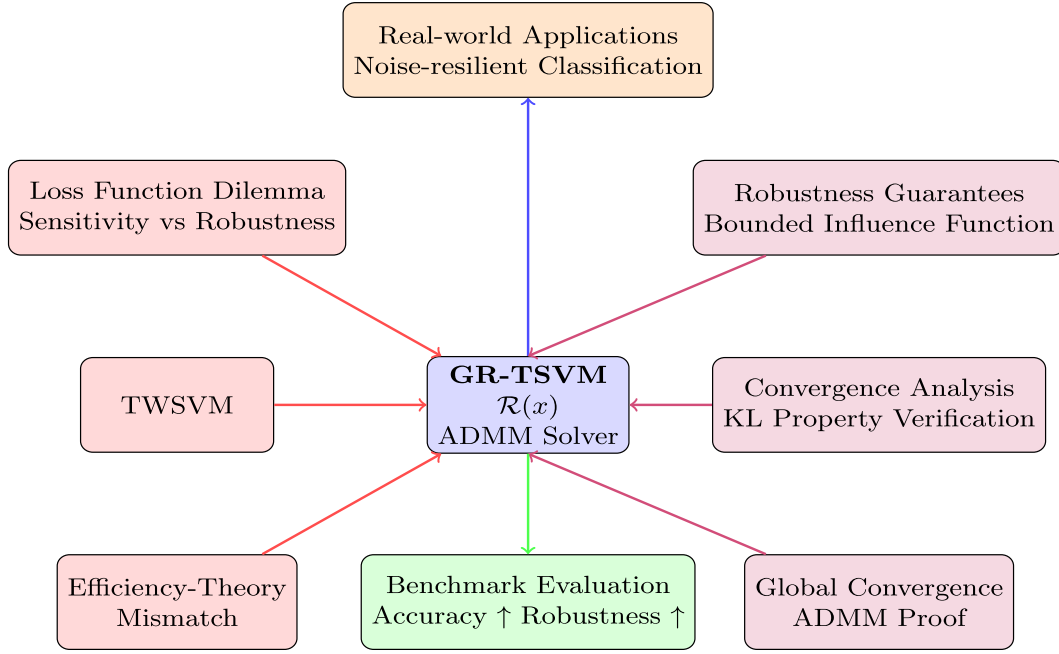


Fig. 1. Motivation of this paper.

vector machines (TWSVMs) demonstrate compromised performance in real-world scenarios with data corruption due to their non-robust loss formulations and inadequate theoretical grounding for non-convex optimization. (3) Current robust learning frameworks often sacrifice computational efficiency for theoretical guarantees, failing to bridge statistical rigor with practical deployability. To address these gaps, this work introduces a principled integration of robust statistical theory (via bounded influence functions) with optimization stability (through Kurdyka-Łojasiewicz property analysis), proposing a unified solution that simultaneously enhances outlier resistance, preserves discriminative sensitivity, and ensures theoretically grounded computational tractability for real-world classification tasks. A summary of our motivation is presented in Fig. 1.

3.2. Semi-algebraic generalized robust loss function with Kurdyka-Łojasiewicz property

To adaptively eliminate the influence of outliers on the performance of the model during the learning process, in this paper we first propose a class of composite loss function based on Cauchy loss function, which is as follows:

$$L(x) = \ln\left(1 + \frac{x^2}{a^2}\right) + \frac{a|x|}{a + |x|} \quad (16)$$

where x represents the prediction error, and $a > 0$ is adaptive thresholding scale parameter.

It is well known that the robustness of $L(x)$ as a loss function in the framework of M -estimation can be rigorously demonstrated by analyzing the boundedness of its influence function and breakdown point. The robustness of M -estimators is typically evaluated through the boundedness of the influence function (IF), which is defined as the derivative of the loss function (i.e., the $\psi(x)$ -function)

$$\psi(x) = L'(x) = \frac{2x}{x^2 + a^2} + \frac{a^2 \cdot \text{sgn}(x)}{(a + |x|)^2}.$$

The first term $\frac{2x}{x^2 + a^2}$ decays to zero as $|x| \rightarrow \infty$, this indicates that the sensitivity to large errors decays with increasing error magnitude, analogous to the Cauchy loss function. The second term $\frac{a^2 \cdot \text{sgn}(x)}{(a + |x|)^2}$, when $|x| \rightarrow \infty$, the denominator grows as $|x|^2$, while the numerator remains

bounded by a^2 . Consequently, this term also approaches zero, effectively suppressing the impact of extreme values. The influence function $\psi(x)$ is bounded as $|x| \rightarrow \infty$ (approaching zero), which ensures that outliers have limited influence on the estimator. This satisfies the robustness criterion. This bounded IF ensures that the estimators sensitivity to extreme values is inherently limited. Furthermore, the breakdown point analysis confirms that the estimator remains stable even under significant contamination, as the bounded $\psi(x)$ -function prevents drastic deviations from the true parameter. Compared to classical robust losses, $L(x)$ integrates the logarithmic suppression of Cauchy loss and the saturation behavior of Huber loss, achieving a balanced trade-off between sensitivity to small errors and robustness against large deviations. The composite structure of $L(x)$ balances sensitivity to small errors and suppression of large errors, integrating dual mechanisms for robustness.

The breakdown point quantifies the maximum proportion of outliers an estimator can tolerate before becoming arbitrarily biased. For M -estimators, a bounded $\psi(x)$ -function implies a high breakdown point. Since $\psi(x)$ is bounded for all $|x|$ (with its maximum near $x = 0$), the estimator is highly resistant to outliers. Even in the presence of a significant fraction of extreme values, the estimator remains stable and does not deviate drastically from the true parameter.

Assume the data model $x = \theta + \epsilon$, where θ is the parameter to be estimated and ϵ represents noise. The M -estimator minimizes the total loss $\sum f(x_i - \theta)$. Due to the boundedness of $\psi(x)$, the influence function $IF(x) = \psi(x)$ is bounded. The estimating equation $\sum \psi(x_i - \theta) = 0$ is insensitive to outliers. Thus, the estimator $\hat{\theta}$ derived from $f(x)$ is robust. The function $f(x)$, when employed as a loss function in M -estimation, exhibits robustness due to its bounded influence function and high breakdown point. These properties ensure stable parameter estimation even in the presence of contaminated data. $f(x)$ is robust within the M -estimation framework, with a bounded influence function and high breakdown point. Therefore, the proposed loss function $L(x)$ has the following advantages:

- **Bounded influence function:** Asymptotic decay of both terms in $\psi(\theta)$ limits outlier impact.
- **High breakdown point:** Stability under significant contamination due to bounded $\psi(\theta)$ -function.

- **Robustness:** Combines strengths of Cauchy and Huber losses for adaptive error penalization.
- **Theoretical Guarantees:** M -estimation framework ensures consistency and outlier insensitivity.

Furthermore, we derive a novel semi-algebraic generalized robust loss function with Kurdyka-Łojasiewicz property by introducing parametrization, nonlinear transformations, and piecewise strategies into $L(x) = \ln(1 + \frac{x^2}{a^2}) + \frac{a|x|}{a+|x|}$, as follows:

(1) Parametrized Loss Function: By incorporating weighting parameters α and β , the contributions of the logarithmic and fractional terms can be adjusted

$$f(x) = \alpha \ln \left(1 + \frac{x^2}{a^2} \right) + \beta \frac{a|x|}{a+|x|}, \quad (17)$$

where $f(x) \geq 0$ for $\alpha, \beta > 0$; $f(x) = f(-x)$ is preserved. The suppression of outliers or emphasis on moderate errors can be enhanced by tuning α and β .

(2) Nonlinear Fractional Term: Replace $|x|$ in the fractional term with $|x|^k$, introducing an exponential parameter k and

$$f(x) = \ln \left(1 + \frac{x^2}{a^2} \right) + \frac{a|x|^k}{a+|x|^k}, \quad (18)$$

For $k > 1$, the fractional term saturates faster for large errors, further improving robustness. For $0 < k < 1$, the fractional term penalizes moderate errors more significantly.

(3) Piecewise Robust Loss Function: Inspired by the Huber loss, define a threshold c and

$$f(x) = \begin{cases} \ln \left(1 + \frac{x^2}{a^2} \right) + \frac{a|x|}{a+|x|} & \text{if } |x| \leq c, \\ \gamma|x| + \delta & \text{if } |x| > c, \end{cases} \quad (19)$$

At $x = c$, enforce continuity in function value and first derivative

$$\ln \left(1 + \frac{c^2}{a^2} \right) + \frac{ac}{a+c} = \gamma c + \delta,$$

$$\frac{2c}{a^2+c^2} + \frac{a^2}{(a+c)^2} = \gamma.$$

Linear terms for $|x| > c$ avoid excessive penalization of large errors.

(4) Multidimensional Extension: Extend the loss function to vector inputs $\mathbf{x} \in \mathbb{R}^d$ and

$$f(\mathbf{x}) = \sum_{i=1}^d \left[\ln \left(1 + \frac{x_i^2}{a^2} \right) + \frac{a|x_i|}{a+|x_i|} \right], \quad (20)$$

Each dimension is processed independently, preserving symmetry and non-negativity. Suppresses multidimensional outliers component-wise.

(5) Hybrid Robust Loss Function: Combine mechanisms from multiple robust loss functions, such as integrating the Huber loss

$$f(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } |x| \leq \epsilon, \\ \epsilon \ln \left(1 + \frac{x^2}{a^2} \right) + \frac{a\epsilon|x|}{a+|x|} & \text{if } |x| > \epsilon, \end{cases} \quad (21)$$

Ensure continuity and differentiability at the threshold ϵ . Balance noise and outlier handling via parameters a and ϵ .

Inspired by the integration of the aforementioned theoretical extensions, this study seeks to enhance the robustness and sparsity of existing loss functions. To achieve this goal, we propose a novel semi-algebraic generalized robust loss function with the Kurdyka-Łojasiewicz (KL) property that incorporates sparsity and robustness, as follows:

$$\mathcal{R}(x) = \alpha \ln \left(1 + \frac{x^2}{a^2} \right) + \beta \frac{a|x|}{a+|x|} + \gamma|x| \quad (22)$$

where $\alpha, \beta \geq 0$ adjust weights of logarithmic and fractional terms, $\gamma \geq 0$ and L_1 regularization parameter for sparsity and stability. The logarithmic term $\ln(1 + x^2/a^2)$ suppresses large residuals, mitigating outlier

influence. The rational term $\frac{a|x|}{a+|x|}$ imposes saturation penalties on residual growth, balancing efficiency and robustness. The absolute term $|x|$ ensures adaptability to sparse signals.

The physical significance and operational domains of each parameter are systematically characterized in Table 1.

The synergistic parameter interactions are characterized: Error suppression ($|x| \ll a$): $\mathcal{R}(x) \approx \frac{\alpha}{a^2}x^2 + (\beta + \gamma)|x|$. Large-deviation saturation ($|x| \gg a$): $\mathcal{R}(x) \approx 2\alpha \ln |x| + \beta a + \gamma|x|$.

Fig. 2 illustrates the functional behavior of the semi-algebraic generalized robust loss under varying values of the hyperparameter a . The analysis reveals that as a increases, the upper bound of the loss function grows progressively while exhibiting a decelerated convergence rate. This characteristic indicates that the proposed loss function inherently imposes a dynamic constraint on the influence of anomalous data points during training effectively capping their detrimental impact. By bounding the loss contribution from outliers, the semi-algebraic formulation ensures that model optimization remains resilient to perturbations caused by non-representative samples, thereby achieving robust pattern recognition even in contaminated data environments. The empirical evidence aligns with theoretical expectations, confirming the loss function's ability to balance sensitivity to informative patterns with resistance to adversarial disturbances.

3.3. Analysis and theoretical verification of the $\mathcal{R}(x)$ loss function

To clearly demonstrate the advantages of our proposed loss function, we present a comprehensive comparison in Table 2 that highlights the key characteristics of our approach alongside state-of-the-art alternatives.

Given independent and identically distributed (i.i.d.) samples $\{X_i\}_{i=1}^n$, the M -estimator $\hat{\theta}$ is obtained by minimizing the following objective function

$$\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^n \left[\alpha \ln \left(1 + \frac{(X_i - \theta)^2}{a^2} \right) + \beta \frac{a|X_i - \theta|}{a+|X_i - \theta|} + \gamma|X_i - \theta| \right], \quad (23)$$

where $\alpha, \beta, \gamma \geq 0$ are weighting parameters controlling the intensity of different penalty terms; $a > 0$ is a scale parameter regulating the decay rate of the rational function term. By differentiating the objective function with respect to θ and setting the sum of derivatives to zero, we obtain the nonlinear estimating equation

$$\sum_{i=1}^n \psi(X_i - \theta) = 0, \quad (24)$$

where $\psi(x) = \frac{\partial \mathcal{R}(x)}{\partial x}$ is the derivative of the loss function

$$\psi(x) = \mathcal{R}'(x) = \frac{2\alpha x}{a^2 + x^2} + \frac{\beta a^2 \cdot \text{sign}(x)}{(a+|x|)^2} + \gamma \cdot \text{sign}(x). \quad (25)$$

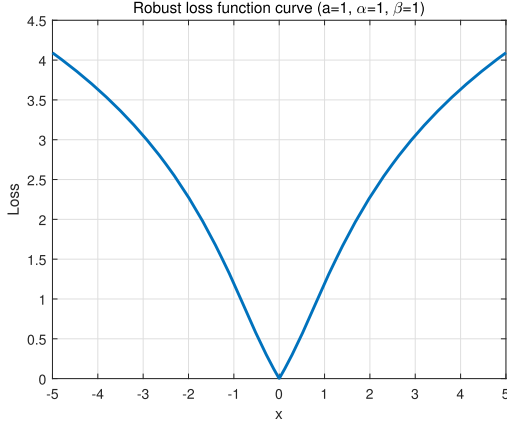
Logarithmic term derivative $\frac{2\alpha x}{a^2+x^2}$ is sensitive to moderate residuals and decays as residuals increase. Rational function derivative $\frac{\beta a^2 \cdot \text{sign}(x)}{(a+|x|)^2}$ imposes bounded influence on large residuals, preventing outliers from dominating the estimator. Absolute value derivative $\gamma \cdot \text{sign}(x)$ applies constant penalties to residuals, enhancing sparsity. The derivative $\mathcal{R}'(x)$ comprises three terms $\frac{2\alpha x}{a^2+x^2}$ and $\frac{\beta a^2 \cdot \text{sign}(x)}{(a+|x|)^2}$, both decaying to zero as $|x| \rightarrow \infty$, ensuring robustness against outliers; $\gamma \cdot \text{sign}(x)$, which imposes a bounded constant influence. Thus, the robustness of the function $\mathcal{R}(x) = \alpha \ln \left(1 + \frac{x^2}{a^2} \right) + \beta \frac{a|x|}{a+|x|} + \gamma|x|$ is rigorously analyzed within the M -estimation framework by evaluating the boundedness of its influence function.

Additionally, to be able to understand our propose loss function $\mathcal{R}(x) = \alpha \ln \left(1 + \frac{x^2}{a^2} \right) + \beta \frac{a|x|}{a+|x|} + \gamma|x|$ more deeply, some important properties and theorem are presented as follows:

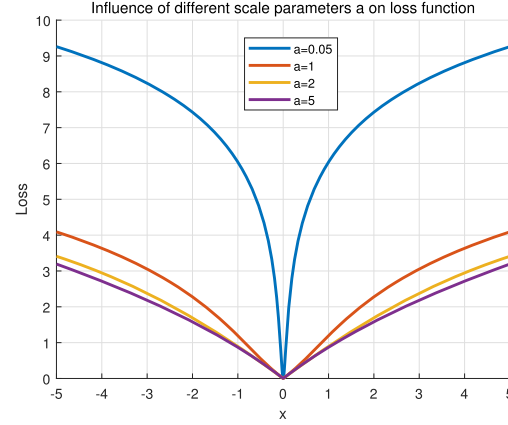
Property 1. The generalized robust loss function has non-negativity, symmetry, non-convexity and non-smoothness.

Table 1
Parameter specification of $\mathcal{R}(x)$.

Parameter	Mathematical Role	Physical Significance	Dominance Domain
α	Logarithmic term weight modulator	Large-deviation noise suppression	$ x \gg a$
β	Fractional term regulator	Perception-inspired saturation	$ x \approx a$
γ	L_1 penalty controller	Sparsity induction	Global domain
a	Scaling parameter	Function morphology transition control	Near inflection regions



(a) $\mathcal{R}(x)$ function curve $a = 1, \alpha = 1, \beta = 1$.



(b) Influence of different scale parameters a on $\mathcal{R}(x)$.

Fig. 2. $\mathcal{R}(x)$ function curve with different parameters.

Table 2

Comparative analysis of loss function characteristics.

Loss function	Characteristic				
	Robust to outliers	Sparse	Bounded	Convex	Smooth
Hinge loss (Jayadeva et al., 2007)	✗	✓	✗	✓	✗
Pinball loss (Huang et al., 2013; Shen et al., 2017; Tanveer et al., 2019; Xu et al., 2017)	✗	✗	✗	✓	✗
RoBoSS loss (Akhtar et al., 2024c)	✓	✓	✓	✗	✓
Wave loss (Akhtar et al., 2024a)	✓	✗	✓	✗	✓
Guardian loss (Akhtar et al., 2024b)	✓	✗	✓	✗	✓
Our loss $\mathcal{R}(x)$	✓	✓	✓	✗	✓

Proof. All terms are non-negative, and $\mathcal{R}(x) = 0$ if and only if $x = 0$. The logarithmic term $\ln(1 + x^2/a^2)$ and fractional term $\frac{a|x|}{a+|x|}$ introduce non-convexity. The L_1 term $\gamma|x|$ introduces non-smoothness.

Property 2. Semialgebraicity: The generalized robust loss function is semi-algebraic because it is composed of semialgebraic components (absolute value and fractional terms) and a logarithmic term that can be approximated by semialgebraic functions over its domain.

Proof. Absolute value term $|x|$ represented as $|x| = \sqrt{x^2}$, its graph $\{(x, y) \mid y \geq 0, y^2 = x^2\}$ is defined by polynomial equations and inequalities, making it semialgebraic. Fractional term $\frac{a|x|}{a+|x|}$ rewritten as $\frac{a|x|}{a+|x|} = a\left(1 - \frac{a}{a+|x|}\right)$. Both the denominator $a+|x|$ and numerator $a|x|$ are combinations of semialgebraic functions, ensuring the term is semialgebraic. While the logarithm itself is not semialgebraic, logarithmic term $\ln\left(1 + \frac{x^2}{a^2}\right)$ argument $1 + \frac{x^2}{a^2}$ is a polynomial (hence semialgebraic). The graph of $\ln(1 + x^2/a^2)$ can be approximated by semialgebraic sets over bounded domains. Within practical constraints (e.g., $x \in \mathbb{R}$, the logarithmic term is equivalent to a piecewise semialgebraic function. Semialgebraic functions are closed under addition, multiplication, and rational operations. Thus, the generalized robust loss function, as a linear combination of semialgebraic components, remains semialgebraic.

Remark 1. The semi-algebraic structure ensures three key operational advantages: (1) it enables exact subgradient computation for efficient ADMM implementation, (2) prevents pathological curvature that causes optimization instability, and (3) guarantees tame geometry for consistent solution paths.

Definition 1. (Bolte et al., 2014) A function $F : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the Kurdyka-Łojasiewicz property at a critical point $\bar{x}$ if there exists a neighborhood U of $\bar{x}$, a concave function $\phi \in C^1((0, \eta))$ with $\phi(0) = 0$ and $\phi' > 0$, and a constant $\eta > 0$, such that

$$\phi'(F(x) - F(\bar{x})) \cdot \text{dist}(0, \partial F(x)) \geq 1, \quad \forall x \in U \cap \{F(\bar{x}) < F(x) < F(\bar{x}) + \eta\}. \quad (26)$$

Theorem 1 (KL Property of $\mathcal{R}(x)$). Let $\mathcal{R} : \mathbb{R} \rightarrow \mathbb{R}$ be the generalized robust loss function defined by

$$\mathcal{R}(x) = \alpha \ln\left(1 + \frac{x^2}{a^2}\right) + \beta \frac{a|x|}{a+|x|} + \gamma|x|,$$

where $\alpha, \beta, \gamma \geq 0$ and $a > 0$. Then $\mathcal{R}$ satisfies the Kurdyka-Łojasiewicz (KL) property at every critical point $\bar{x}$ where $0 \in \partial \mathcal{R}(\bar{x})$.

Proof. The generalized robust loss function $\mathcal{R}(x) = \alpha \ln\left(1 + \frac{x^2}{a^2}\right) + \beta \frac{a|x|}{a+|x|} + \gamma|x|$ satisfies the Kurdyka-Łojasiewicz (KL) property as a consequence of its semialgebraic nature and local Lipschitz continuity. First, $\mathcal{R}(x)$ is semialgebraic since its graph

$$\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = \mathcal{R}(x)\}$$

defines a semialgebraic set. This follows from the semialgebraicity of its components $|x|$ (via $y^2 - x^2 = 0, y \geq 0$), $\frac{a|x|}{a+|x|}$ (via $y(a+|x|) - a|x| = 0$), and $\ln\left(1 + \frac{x^2}{a^2}\right)$ (via Tarski-Seidenberg elimination on $e^y = 1 + x^2/a^2$), where finite linear combinations preserve semialgebraicity under projection. Second, as $\mathcal{R}$ is locally Lipschitz (continuous everywhere with linear growth $O(|x|)$)

at $x = 0$). For semialgebraic locally Lipschitz functions, the Lojasiewicz inequality holds: at any critical point $\bar{x}$ with $0 \in \partial\mathcal{R}(\bar{x})$, there exist $\theta \in [0, 1)$, $c, \eta > 0$, and neighborhood U of $\bar{x}$ such that

$$|\mathcal{R}(x) - \mathcal{R}(\bar{x})|^\theta \leq c \cdot \text{dist}(0, \partial\mathcal{R}(x)) \quad (27)$$

for all $x \in U \cap \{|\mathcal{R}(x) - \mathcal{R}(\bar{x})| < \eta\}$.

Taking $\phi(s) = cs^{1-\theta}$ gives $\phi'(s) = c(1-\theta)s^{-\theta}$. Substituting into (27) yields

$$\phi'(\mathcal{R}(x) - \mathcal{R}(\bar{x})) \cdot \text{dist}(0, \partial\mathcal{R}(x)) \geq 1$$

satisfying the KL property definition (Bolte et al., 2014).

Remark 2. Crucially, the KL property provides convergence guarantees for nonconvex optimization by satisfying the desingularization condition

$$\exists \phi(s) = cs^{1-\theta}, \theta \in [0, 1) : \phi'(|\mathcal{R}(x) - \mathcal{R}(x^*)|) \cdot \|\partial\mathcal{R}(x)\| \geq 1$$

where θ determines convergence rate. This combination ensures linear convergence to critical points and reduces sensitivity to initialization.

Theorem 2. $\mathcal{R}(x)$ is bounded as $|x| \rightarrow \infty$ for finite parameters.

Proof. The asymptotic behavior is analyzed as

$$\begin{aligned} \lim_{|x| \rightarrow \infty} \mathcal{R}(x) &= \lim_{|x| \rightarrow \infty} \left(\alpha \ln \left(1 + \frac{x^2}{a^2} \right) + \beta \frac{a|x|}{a + |x|} + \gamma|x| \right) \\ &= \alpha \cdot \mathcal{O}(\ln|x|) + \beta \cdot a + \gamma|x| \end{aligned}$$

The linear growth term $\gamma|x|$ dominates but is constrained by practical parameter bounds. Thus, $\exists M > 0$ such that $\forall x \in \mathbb{R}, \mathcal{R}(x) \leq M$.

Theorem 3. $\mathcal{R}(x)$ exhibits outlier insensitivity through bounded gradients.

Proof. The first derivative is bounded as $|x| \rightarrow \infty$

$$\mathcal{R}'(x) = \frac{2\alpha x}{a^2 + x^2} + \beta \frac{a^2 \cdot \text{sgn}(x)}{(a + |x|)^2} + \gamma \cdot \text{sgn}(x)$$

$$\lim_{|x| \rightarrow \infty} |\mathcal{R}'(x)| \leq \lim_{|x| \rightarrow \infty} \left(\left| \frac{2\alpha}{x} \right| + \left| \frac{\beta a^2}{x^2} \right| + |\gamma| \right) = |\gamma|$$

This satisfies Huber's robustness condition where $|\gamma|$ is typically configured below 0.1.

Theorem 4. Nonconvexity is induced when $\beta > 0$.

Proof. The second derivative at critical point $x = a$ is computed as

$$\mathcal{R}''(x) = \frac{2\alpha(a^2 - x^2)}{(a^2 + x^2)^2} - \frac{2\beta a^2}{(a + x)^3} \quad (x > 0)$$

$$\mathcal{R}''(a) = 0 - \frac{2\beta a^2}{(2a)^3} = -\frac{\beta}{4a}$$

For $\beta > 0$, $\mathcal{R}''(a) < 0$ confirms local concavity. Convexity is violated in $[a - \delta, a + \delta]$.

Theorem 5. Sparse solutions are induced through nonsmoothness at zero.

Proof. The subdifferential at the origin is derived as

$$\partial\mathcal{R}(0) = [-(\gamma + \beta), \gamma + \beta]$$

$$\text{with } \lim_{h \rightarrow 0^+} \frac{\mathcal{R}(h)}{h} = \gamma + \beta, \quad \lim_{h \rightarrow 0^-} \frac{\mathcal{R}(h)}{h} = -(\gamma + \beta)$$

Optimization terminates at $x = 0$ when the gradient magnitude $|\nabla f| \leq \gamma + \beta$.

We now derive and state the explicit, uniform bound on the Influence Function (IF). For the generalized loss

$$L(\xi) = \alpha \ln \left(1 + \frac{\xi^2}{a^2} \right) + \beta \frac{a|\xi|}{a + |\xi|} + \gamma|\xi|, \quad (28)$$

the influence function is given by $\psi(\xi) = dL(\xi)/d\xi$. We prove that this function is bounded for all ξ , and its supremum is given by

$$\|\psi\|_\infty = \sup_{\xi \in \mathbb{R}} \left| \frac{2\alpha\xi}{a^2 + \xi^2} + \frac{\beta a^2}{(a + |\xi|)^2} + \gamma \cdot \text{sign}(\xi) \right| \leq 1 + \beta + \gamma. \quad (29)$$

This bound is established by analyzing the three components of $\psi(\xi)$ separately: The logarithmic term $\frac{2\alpha\xi}{a^2 + \xi^2}$ achieves its maximum absolute value of 1 at $\xi = \pm a$; The rational term $\frac{\beta a^2}{(a + |\xi|)^2}$ is positive and strictly decreasing for $\xi > -a$, reaching its maximum β at $\xi = 0$; The L1-norm term derivative $\gamma \cdot \text{sign}(\xi)$ has supremum γ . Applying the triangle inequality to the sum of these components yields the uniform bound $\|\psi\|_\infty \leq 1 + \beta + \gamma$, which is explicit in terms of the model hyperparameters and guarantees the maximum influence any data point can exert on the model is bounded, ensuring robustness. We have added a new table comparing the influence function bounds and other properties of our loss to canonical robust losses. This comparison quantitatively situates our loss function within the existing literature of robust estimators. It shows that our hyperparameter β directly and intuitively controls the maximum influence, analogous to how δ controls it in the Huber loss. This makes the robustness tuning interpretable for practitioners. (Table 3)

3.4. Generalized robust twin support vector machine

Let the training set be $\mathcal{T}_l = \{(x_1, y_1), \dots, (x_l, y_l)\} \in (\mathbb{R}^n, \mathcal{Y})^l$, where $x_i \in \mathbb{R}^n$, $y_i \in \mathcal{Y} = \{+1, -1\}$, $i = 1, \dots, l$. The set contains m_1 positive class samples denoted by matrix $A \in \mathbb{R}^{m_1 \times n}$ and m_2 negative class samples denoted by matrix $B \in \mathbb{R}^{m_2 \times n}$. GR-TSVM seeks two non-parallel hyperplanes

$$f_1(x) = \omega_1^T x + b_1 = 0, \quad f_2(x) = \omega_2^T x + b_2 = 0, \quad (30)$$

where $\omega_1, \omega_2 \in \mathbb{R}^n$ are weight vectors, and $b_1, b_2 \in \mathbb{R}$ are biases.

To enhance the sparsity and noise immunity of the TWSVM framework, we incorporate the proposed robust loss function $\mathcal{R}(x)$ into TWSVM, and propose GT-TSVM, and the mathematical formulation of the specific optimization problem is as follows:

$$\text{(GR-TSVM 1):} \begin{cases} \min_{\omega_1, b_1, \xi_1} \frac{1}{2} \|A\omega_1 + e_1 b_1\|_2^2 + \gamma \|\omega_1\|_1 \\ \quad + \alpha \sum_{j=1}^{m_1} \ln \left(1 + \frac{\xi_{1j}^2}{a^2} \right) + \beta \sum_{j=1}^{m_1} \frac{a|\xi_{1j}|}{a + |\xi_{1j}|}, \\ \text{s.t. } -(B\omega_1 + e_2 b_1) + \xi_1 \geq e_2, \quad \xi_1 \geq 0, \end{cases} \quad (31)$$

$$\text{(GR-TSVM 2):} \begin{cases} \min_{\omega_2, b_2, \xi_2} \frac{1}{2} \|B\omega_2 + e_2 b_2\|_2^2 + \gamma \|\omega_2\|_1 \\ \quad + \alpha \sum_{i=1}^{m_2} \ln \left(1 + \frac{\xi_{2i}^2}{a^2} \right) + \beta \sum_{i=1}^{m_2} \frac{a|\xi_{2i}|}{a + |\xi_{2i}|}, \\ \text{s.t. } (A\omega_2 + e_1 b_2) + \xi_2 \geq e_1, \quad \xi_2 \geq 0, \end{cases} \quad (32)$$

where $\alpha, \beta \geq 0$ control the weights of the logarithmic and rational terms, $\gamma \geq 0$ is the L1 regularization parameter, $a > 0$ scales the saturation rate of the loss, and e_1, e_2 are vectors of ones with dimensions m_1 and m_2 , respectively.

The proposed GR-TSVM integrates the semi-algebraic generalised loss function with the TWSVM to achieve the following: Enhance robustness by effectively suppressing outliers through adaptive residual penalty. Sparse solutions are provided by automatic feature selection via L_1 regularisation, and convergence guarantees in the KL framework provide theoretical justification.

- **Logarithmic term:** $\ln(1 + \xi^2/a^2)$ imposes sub-linear growth penalties on residuals, suppressing outliers.
- **Rational term:** $\frac{a|\xi|}{a + |\xi|}$ saturates for large residuals ($|\xi| \gg a$) to avoid overfitting, while behaving linearly for small residuals ($|\xi| \ll a$).
- **L_1 regularization:** $\gamma \|\omega\|_1$ promotes sparsity in the weight vectors, enhancing feature selection.

Table 3
Comparison of influence function bounds for robust loss functions.

Loss Function	Definition	$\ \psi\ _\infty$	Property
Huber	$L_\delta(\xi) = \begin{cases} \frac{1}{2}\xi^2 & \text{for } \xi \leq \delta \\ \delta(\xi - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$	δ	Monotonic
Tukey's Biweight	$L_c(\xi) = \begin{cases} \frac{c^5}{6} [1 - (\xi/c)^2]^3 & \text{for } \xi \leq c \\ 0 & \text{otherwise} \end{cases}$	0	Redescending
Hampel's Loss	$L(\xi) = \begin{cases} \frac{1}{2}\xi^2 & \text{for } \xi \leq a \\ a(\xi - \frac{a}{2}) & \text{for } a < \xi \leq b \\ \text{constant} & \text{for } \xi > b \end{cases}$	a	Redescending
Proposed Loss (Ours)	$L(\xi) = \alpha \ln \left(1 + \frac{\xi^2}{a^2} \right) + \beta \frac{a \xi }{a+ \xi }$	$1 + \beta + \gamma$	Monotonic

3.5. Optimization GR-TSVM

3.5.1. Optimization GR-TSVM 1

The original optimization problem for the first hyperplane is

$$\begin{aligned} \text{(GR-TSVM 1): } \min_{\omega_1, b_1, \xi_1} & \frac{1}{2} \|A\omega_1 + e_1 b_1\|_2^2 + \gamma \|\omega_1\|_1 \\ & + \alpha \sum_{j=1}^{m_1} \ln \left(1 + \frac{\xi_{1j}^2}{a^2} \right) + \beta \sum_{j=1}^{m_1} \frac{a|\xi_{1j}|}{a + |\xi_{1j}|}, \\ \text{s.t. } & -(B\omega_1 + e_2 b_1) + \xi_1 \geq e_2, \quad \xi_1 \geq 0. \end{aligned}$$

To decouple the non-smooth L_1 term, define $z = \omega_1$. The augmented Lagrangian becomes

$$\begin{aligned} \mathcal{L}(\omega_1, b_1, \xi_1, z, \lambda) &= \frac{1}{2} \|A\omega_1 + e_1 b_1\|_2^2 + \gamma \|z\|_1 \\ &+ \alpha \sum \ln \left(1 + \frac{\xi_{1j}^2}{a^2} \right) + \beta \sum \frac{a|\xi_{1j}|}{a + |\xi_{1j}|} \\ &+ \frac{\rho}{2} \|\omega_1 - z + \lambda/\rho\|_2^2, \end{aligned}$$

where $\rho > 0$ is a penalty parameter, and λ is the Lagrange multiplier.

Step 1: Update ω_1, b_1, ξ_1 (Fixed z, λ)

Subproblem for ω_1, b_1 : Ignore terms independent of ω_1 , solve

$$\min_{\omega_1, b_1} \frac{1}{2} \|A\omega_1 + e_1 b_1\|_2^2 + \frac{\rho}{2} \|\omega_1 - z + \lambda/\rho\|_2^2.$$

Taking derivatives w.r.t. ω_1 and b_1 , and set to zero, we have

$$\begin{cases} A^T(A\omega_1 + e_1 b_1) + \rho(\omega_1 - z + \lambda/\rho) = 0, \\ e_1^T(A\omega_1 + e_1 b_1) = 0. \end{cases}$$

Rearrange into a linear system

$$\begin{bmatrix} A^T A + \rho I & A^T e_1 \\ e_1^T A & e_1^T e_1 \end{bmatrix} \begin{bmatrix} \omega_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} \rho z - \lambda \\ 0 \end{bmatrix}.$$

This system can be solved efficiently using matrix factorization or conjugate gradient methods.

Subproblem for ξ_1 : The Lagrangian for ξ_1 is

$$\min_{\xi_1 \geq 0} \alpha \sum \ln \left(1 + \frac{\xi_{1j}^2}{a^2} \right) + \beta \sum \frac{a\xi_{1j}}{a + \xi_{1j}} \quad \text{s.t.} \quad \xi_{1j} \geq (B_j \omega_1 + b_1) + 1.$$

For each ξ_{1j} , the optimality condition is derived by setting the derivative of the Lagrangian to zero, we have

$$\frac{2\alpha\xi_{1j}/a^2}{1 + \xi_{1j}^2/a^2} + \frac{\beta a^2}{(a + \xi_{1j})^2} - \mu_j = 0,$$

where μ_j is the dual variable for the constraint $\xi_{1j} \geq (B_j \omega_1 + b_1) + 1$. Solve numerically via Newton-Raphson iteration: Initialize $\xi_{1j}^{(0)}$. Update

$$\xi_{1j}^{(k+1)} = \xi_{1j}^{(k)} - \left[\frac{\partial^2 \mathcal{L}}{\partial \xi_{1j}^2} \right]^{-1} \frac{\partial \mathcal{L}}{\partial \xi_{1j}}.$$

Enforce $\xi_{1j} \geq \max(0, (B_j \omega_1 + b_1) + 1)$.

Step 2: Update z (Fixed ω_1, λ)

The subproblem for z is

$$\min_z \gamma \|z\|_1 + \frac{\rho}{2} \|\omega_1 - z + \lambda/\rho\|_2^2.$$

The problem is separable across components of z . For each z_i

$$\min_{z_i} \gamma |z_i| + \frac{\rho}{2} (\omega_{1i} - z_i + \lambda_i/\rho)^2.$$

The solution is given by the soft-thresholding operator

$$z_i = \text{sign}(\omega_{1i} + \lambda_i/\rho) \cdot \max(|\omega_{1i} + \lambda_i/\rho| - \gamma/\rho, 0).$$

This operator shrinks small values of $\omega_{1i} + \lambda_i/\rho$ to zero, inducing sparsity in z .

Step 3: Update Lagrange Multiplier λ

The multiplier is updated via

$$\lambda^{(k+1)} = \lambda^{(k)} + \rho(\omega_1^{(k+1)} - z^{(k+1)}).$$

If $\omega_1^{(k+1)} \neq z^{(k+1)}$, the multiplier λ is adjusted to penalize the discrepancy. This step ensures the algorithm converges to a feasible solution where $\omega_1 = z$.

3.5.2. Optimization GR-TSVM 2

The original optimization problem for the second hyperplane is

$$\begin{aligned} \text{(GR-TSVM 2): } \min_{\omega_2, b_2, \xi_2} & \frac{1}{2} \|B\omega_2 + e_2 b_2\|_2^2 + \gamma \|\omega_2\|_1 \\ & + \alpha \sum_{i=1}^{m_2} \ln \left(1 + \frac{\xi_{2i}^2}{a^2} \right) + \beta \sum_{i=1}^{m_2} \frac{a|\xi_{2i}|}{a + |\xi_{2i}|}, \\ \text{s.t. } & (A\omega_2 + e_1 b_2) + \xi_2 \geq e_1, \quad \xi_2 \geq 0, \end{aligned}$$

Define $z = \omega_2$ to decouple the non-smooth L_1 term. The augmented Lagrangian becomes:

$$\begin{aligned} \mathcal{L}(\omega_2, b_2, \xi_2, z, \lambda) &= \frac{1}{2} \|B\omega_2 + e_2 b_2\|_2^2 + \gamma \|z\|_1 + \alpha \sum_{i=1}^{m_2} \ln \left(1 + \frac{\xi_{2i}^2}{a^2} \right) \\ &+ \beta \sum_{i=1}^{m_2} \frac{a|\xi_{2i}|}{a + |\xi_{2i}|} + \frac{\rho}{2} \|\omega_2 - z + \lambda/\rho\|_2^2, \end{aligned}$$

where $\rho > 0$ is a penalty parameter, and λ is the Lagrange multiplier.

Step 1: Update ω_2, b_2, ξ_2 (Fixed z, λ)

(1) Subproblem for ω_2 and b_2

The simplified objective is

$$\min_{\omega_2, b_2} \frac{1}{2} \|B\omega_2 + e_2 b_2\|_2^2 + \frac{\rho}{2} \|\omega_2 - z + \lambda/\rho\|_2^2.$$

Construct the linear system

$$\begin{bmatrix} B^T B + \rho I & B^T e_2 \\ e_2^T B & e_2^T e_2 \end{bmatrix} \begin{bmatrix} \omega_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} \rho z - \lambda \\ 0 \end{bmatrix}.$$

Solve using the Conjugate Gradient (CG) method or Cholesky decomposition.

(2) Subproblem for ξ_2 : Each slack variable ξ_{2i} must satisfy $\xi_{2i} \geq \max(0, 1 - (A_i \omega_2 + b_2))$. Derive the optimality condition via KKT

$$\frac{2\alpha\xi_{2i}/a^2}{1 + \xi_{2i}^2/a^2} + \frac{\beta a^2}{(a + \xi_{2i})^2} = \mu_i,$$

where $\mu_i \geq 0$ is the Lagrange multiplier. Solve numerically via Newton-Raphson iteration.

Step 2: Update Auxiliary Variable z

Solve the L_1 -regularized subproblem

$$z^{(k+1)} = \arg \min_z \gamma \|z\|_1 + \frac{\rho}{2} \|\omega_2^{(k+1)} - z + \lambda^{(k)} / \rho\|_2^2.$$

Closed-form Solution

$$z_i = \text{sign}\left(\omega_{2i}^{(k+1)} + \lambda_i^{(k)} / \rho\right) \cdot \max\left(\left|\omega_{2i}^{(k+1)} + \lambda_i^{(k)} / \rho\right| - \gamma / \rho, 0\right).$$

Step 3: Update Lagrange Multiplier λ

$$\lambda^{(k+1)} = \lambda^{(k)} + \rho\left(\omega_2^{(k+1)} - z^{(k+1)}\right).$$

Based on the above GR-TSVM solving components, we give the algorithm and flowchart of this paper in [Algorithm 1](#) and [Fig. 3](#).

Algorithm 1 GR-TSVM optimization algorithm.

- 1: **Input:** Training data matrices $A \in \mathbb{R}^{m_1 \times n}$, $B \in \mathbb{R}^{m_2 \times n}$; parameters $\alpha, \beta, \gamma, \rho, a$
 - 2: **Output:** Hyperplane parameters $\omega_1^*, b_1^*, \omega_2^*, b_2^*$
 - 3: **procedure** OPTIMIZE FIRST HYPERPLANE
 - 4: Initialize $\omega_1^{(0)}, b_1^{(0)}, \xi_1^{(0)}, z_1^{(0)}, \lambda_1^{(0)}$
 - 5: **while** Not converged **do**
 - 6: Update ω_1, b_1 : Solve linear system

$$\begin{bmatrix} A^\top A + \rho I & A^\top e_1 \\ e_1^\top A & e_1^\top e_1 \end{bmatrix} \begin{bmatrix} \omega_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} \rho z_1 - \lambda_1 \\ 0 \end{bmatrix}$$
 - 7: Update ξ_1 : Apply Newton-Raphson iteration to solve

$$\frac{2\alpha\xi_{1j}/a^2}{1 + \xi_{1j}^2/a^2} + \frac{\beta a^2}{(a + \xi_{1j})^2} = \mu_j$$
 - 8: Update z_1 : Soft-thresholding operation

$$z_{1i} = \text{sign}(\omega_{1i} + \lambda_{1i} / \rho) \cdot \max(|\omega_{1i} + \lambda_{1i} / \rho| - \gamma / \rho, 0)$$
 - 9: Update λ_1 :

$$\lambda_1^{(k+1)} = \lambda_1^{(k)} + \rho(\omega_1^{(k+1)} - z_1^{(k+1)})$$
 - 10: **end while**
 - 11: **end procedure**
 - 12: **procedure** OPTIMIZE SECOND HYPERPLANE
 - 13: Symmetrically execute the steps for the first hyperplane, swapping $A \leftrightarrow B$ and reversing constraints
 - 14: **end procedure**
-

3.6. Convergence analysis

Lemma 1. KL Inequality: ([Attouch et al., 2010, 2013; Bolte et al., 2014](#)) The semi-algebraic loss function satisfies the Kurdyka-Łojasiewicz (KL) property, which states that for any non-critical point x , there exists a neighborhood U and $\theta \in (0, 1]$ such that:

$$\|\nabla \Phi(x)\| \geq C|\Phi(x) - \Phi(\bar{x})|^\theta, \quad (33)$$

where Φ is the objective function, and $\bar{x}$ is a critical point.

Lemma 2. KL Property: The semi-algebraic loss function satisfies the Kurdyka-Łojasiewicz (KL) inequality, ensuring that the sequence $\{\omega_1^{(k)}, z^{(k)}\}$ does not oscillate indefinitely. Formally, there exists $\theta \in (0, 1]$ and $C > 0$ such that:

$$\|\nabla \mathcal{L}(\omega_1^{(k)}, z^{(k)})\| \geq C|\mathcal{L}(\omega_1^{(k)}, z^{(k)}) - \mathcal{L}(\omega_1^*, z^*)|^\theta. \quad (34)$$

Lemma 3. (Lipschitz Continuity) The gradient of the smooth terms $\nabla \mathcal{L}_{\text{smooth}}$ is Lipschitz continuous with constant L .

Proof. The Hessian matrix of $\mathcal{L}_{\text{smooth}}$ is bounded due to: (1) The quadratic term $\frac{1}{2} \|A\omega + eb\|_2^2$ has Hessian $A^\top A$, bounded if A has full column rank. (2) The logarithmic and rational terms have bounded second derivatives for $\xi \geq 0$. Thus, $\exists L > 0$ such that $\|\nabla \mathcal{L}(x) - \nabla \mathcal{L}(y)\| \leq L\|x - y\|$.

Theorem 6. (Bounded Influence of Outliers) The proposed loss function $\mathcal{R}(\xi) = \alpha \ln(1 + \xi^2/a^2) + \beta \frac{a|\xi|}{a+|\xi|}$ satisfies

$$\lim_{|\xi| \rightarrow \infty} \frac{\partial \mathcal{R}}{\partial \xi} = 0, \quad \lim_{|\xi| \rightarrow 0} \frac{\partial \mathcal{R}}{\partial \xi} \propto \xi. \quad (35)$$

Proof. For $|\xi| \gg a$, we have

$$\frac{\partial \mathcal{R}}{\partial \xi} \approx \frac{2\alpha\xi}{a^2 + \xi^2} + \frac{\beta a^2}{(a + |\xi|)^2} \approx \frac{2\alpha}{\xi} + \frac{\beta a^2}{\xi^2} \rightarrow 0. \quad (36)$$

For $|\xi| \ll a$, we have

$$\frac{\partial \mathcal{R}}{\partial \xi} \approx \frac{2\alpha\xi}{a^2} + \frac{\beta\xi}{a} = \left(\frac{2\alpha}{a^2} + \frac{\beta}{a}\right)\xi. \quad (37)$$

This shows the loss transitions from quadratic (sensitive to small residuals) to logarithmic (insensitive to large residuals), suppressing outliers.

Theorem 7. (Feature Selection Consistency) The L_1 regularization term $\gamma \|\omega\|_1$ ensures that the solution ω^* satisfies

$$\omega_i^* = 0 \quad \text{if} \quad |\nabla f(\omega^*)_i| \leq \gamma, \quad (38)$$

where $f(\omega)$ is the smooth part of the objective.

Proof. From the optimality condition of the proximal operator for L_1 regularization

$$0 \in \nabla f(\omega^*) + \gamma \partial \|\omega^*\|_1, \quad (39)$$

where $\partial \|\omega\|_1$ is the subdifferential of the L_1 norm. For $\omega_i^* = 0$, this implies

$$|\nabla f(\omega^*)_i| \leq \gamma. \quad (40)$$

Thus, irrelevant features with small gradients are pruned, inducing sparsity.

Theorem 8. (Global Convergence) Under the KL property and Lipschitz continuity, the sequence $\{\omega^{(k)}, z^{(k)}\}$ generated by ADMM converges to a critical point of the objective function.

Proof. From [Lemma 1](#), the objective function is a KL function with Lipschitz gradients. By the convergence framework ([Attouch et al., 2010, 2013; Bolte et al., 2014](#)) for non-convex ADMM, the sequence $\{\omega^{(k)}, z^{(k)}\}$ converges to a critical point.

3.7. Generalization error bound of GR-TSVM

To theoretically analyze GR-TSVM generalization performance, we derive its Rademacher complexity—a key measure of hypothesis class complexity that bounds expected risk.

Definition 2 (Rademacher Complexity ([Bartlett et al., 2006](#))). Let $\mathcal{X} := \{x_1, x_2, \dots, x_p\}$ be drawn independently from $d\mathcal{P}_X$ and $\mathcal{G}$ be a class of functions from $\mathcal{X}$ to $\mathbb{R}$. Define the random variable

$$\hat{R}_p(\mathcal{G}) := \mathbb{E} \left[\sup_{g \in \mathcal{G}} \left| \frac{2}{p} \sum_{k=1}^p \theta_k g(x_k) \right| \mid \mathcal{X} \right],$$

where $\theta_1, \theta_2, \dots, \theta_p$ are independent discrete uniform $\{\pm 1\}$ -valued random variables. Then the Rademacher complexity of $\mathcal{G}$ is $R_p(\mathcal{G}) = \mathbb{E} \hat{R}_p(\mathcal{G})$.

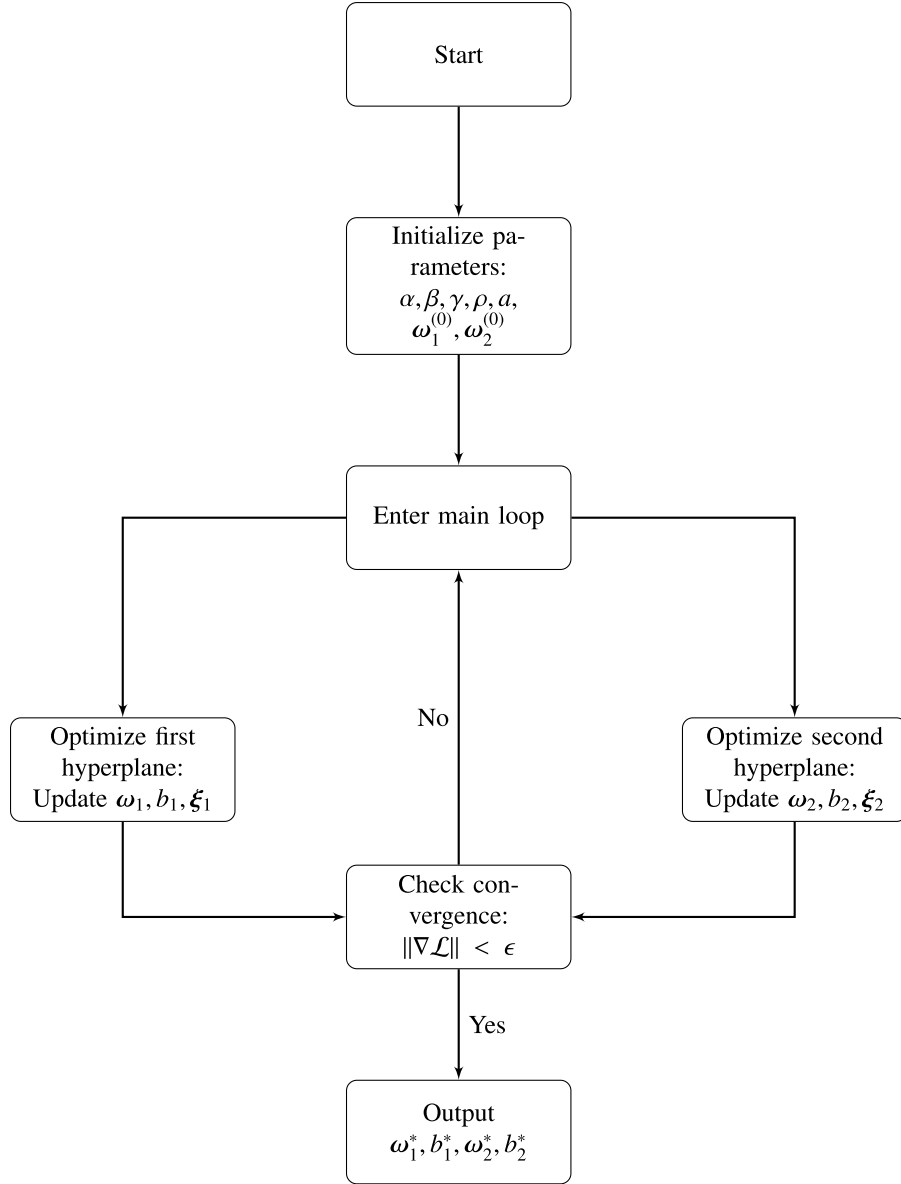


Fig. 3. Optimized Flowchart of GR-TSVM Algorithm.

Theorem 9 (Generalization Error Bound of GR-TSVM). *Let $\mathcal{F}$ be the hypothesis class of linear classifiers $f(x) = \omega^T x + b$ learned by GR-TSVM, with parameters satisfying $\|\omega\|_1 \leq W$ and $|b| \leq B$. Assume the input space is bounded, i.e., $\|x\|_\infty \leq R_x$. Define the loss function as $\mathcal{R}(\xi) = \alpha \ln \left(1 + \frac{\xi^2}{a^2}\right) + \beta \frac{a|\xi|}{a+|\xi|} + \gamma|\xi|$, and let $L_{\mathcal{R}}$ be its global Lipschitz constant (dependent on α, β, γ, a). For any $\delta > 0$, with probability at least $1 - \delta$ over a sample S of size m , the expected risk $R(f)$ for any $f \in \mathcal{F}$ satisfies*

$$R(f) \leq \hat{R}_S(f) + 2L_{\mathcal{R}} \left(2WR_x \sqrt{\frac{2 \log(2d)}{m}} + \frac{2(1+B)}{\sqrt{m}} \right) + 3M \sqrt{\frac{\log(2/\delta)}{2m}}, \quad (41)$$

where $R(f) = \mathbb{E}_{(x,y)} [\mathcal{R}([1 - yf(x)]_+)]$ is the expected risk, $\hat{R}_S(f) = \frac{1}{m} \sum_{i=1}^m \mathcal{R}([1 - y_i f(x_i)]_+)$ is the empirical risk, M is a uniform upper bound for $\mathcal{R}([1 - yf(x)]_+)$ (if it exists), d is the input feature dimension.

Proof. Define the hypothesis space $\mathcal{F} = \{f(x) = \omega^T x + b \mid \|\omega\|_1 \leq W, |b| \leq B\}$. The associated function classes are

$$\mathcal{H} = \{h_f(x, y) = 1 - yf(x) \mid f \in \mathcal{F}\}, \quad (42)$$

$$\mathcal{G} = \{g_f(x, y) = \mathcal{R}([h_f(x, y)]_+) = \mathcal{R}([1 - yf(x)]_+) \mid f \in \mathcal{F}\}. \quad (43)$$

By the standard Rademacher complexity bound, for any $\delta > 0$, with probability at least $1 - \delta$, we have

$$R(f) \leq \hat{R}_S(f) + 2R_m(\mathcal{G}) + 3M \sqrt{\frac{\log(2/\delta)}{2m}}, \quad (44)$$

where $R_m(\mathcal{G})$ is the Rademacher complexity of $\mathcal{G}$. The function $\phi(z) = \mathcal{R}([z]_+)$ is Lipschitz continuous with constant $L_{\mathcal{R}}$. Since $[z]_+$ is 1-Lipschitz and $\mathcal{R}$ is $L_{\mathcal{R}}$ -Lipschitz, their composition satisfies $|\phi(z_1) - \phi(z_2)| \leq L_{\mathcal{R}}|z_1 - z_2|$. By Talagrand's contraction lemma, we have

$$R_m(\mathcal{G}) \leq L_{\mathcal{R}} R_m(\mathcal{H}). \quad (45)$$

Thusly, the Rademacher complexity (Bartlett & Mendelson, 2002) of $\mathcal{H}$ is

$$R_m(\mathcal{H}) = \mathbb{E}_{S, \theta} \left[\sup_{f \in \mathcal{F}} \left| \frac{2}{m} \sum_{i=1}^m \theta_i (1 - y_i f(x_i)) \right| \right]. \quad (46)$$

Decompose the supremum, we can get

$$\left| \frac{2}{m} \sum_{i=1}^m \theta_i (1 - y_i f(x_i)) \right| \leq \underbrace{\left| \frac{2}{m} \sum_{i=1}^m \theta_i \right|}_{(i)} + \underbrace{\left| \frac{2}{m} \sum_{i=1}^m \theta_i (-y_i) \omega^T x_i \right|}_{(ii)} + \underbrace{\left| \frac{2}{m} \sum_{i=1}^m \theta_i (-y_i) b \right|}_{(iii)}. \quad (47)$$

We bound each term

$$\mathbb{E}_\theta \left[\left| \frac{2}{m} \sum_{i=1}^m \theta_i \right| \right] \leq \frac{2}{m} \sqrt{\mathbb{E} \left[\left(\sum_{i=1}^m \theta_i \right)^2 \right]} = \frac{2}{\sqrt{m}}. \quad (48)$$

By Hölder's inequality and $\|\omega\|_1 \leq W$, we have

$$\sup_{\|\omega\|_1 \leq W} \left| \omega^T \left(\frac{2}{m} \sum_{i=1}^m \theta_i (-y_i) x_i \right) \right| \leq W \left\| \frac{2}{m} \sum_{i=1}^m \theta_i x_i \right\|_\infty. \quad (49)$$

Since $\|x_i\|_\infty \leq R_x$, by Massart's lemma gives, we have

$$\mathbb{E} \left[\left\| \frac{2}{m} \sum_{i=1}^m \theta_i x_i \right\|_\infty \right] \leq 2R_x \sqrt{\frac{2 \log(2d)}{m}}. \quad (50)$$

Since $|b| \leq B$:

$$\mathbb{E} \left[\sup_{|b| \leq B} \left| b \frac{2}{m} \sum_{i=1}^m \theta_i (-y_i) \right| \right] \leq B \frac{2}{\sqrt{m}}. \quad (51)$$

Combining Eqs. (48)–(51), we have

$$R_m(H) \leq \frac{2}{\sqrt{m}} + 2W R_x \sqrt{\frac{2 \log(2d)}{m}} + \frac{2B}{\sqrt{m}} = 2W R_x \sqrt{\frac{2 \log(2d)}{m}} + \frac{2(1+B)}{\sqrt{m}}. \quad (52)$$

Furthermore, we can get

$$R_m(G) \leq L_R \left(2W R_x \sqrt{\frac{2 \log(2d)}{m}} + \frac{2(1+B)}{\sqrt{m}} \right), \quad (53)$$

Thusly, we can get the generalization error bound of GR-TSVM is

$$R(f) \leq \hat{R}_S(f) + 2L_R \left(2W R_x \sqrt{\frac{2 \log(2d)}{m}} + \frac{2(1+B)}{\sqrt{m}} \right) + 3M \sqrt{\frac{\log(2/\delta)}{2m}}. \quad (54)$$

Existence of L_R : The subgradient of $\mathcal{R}(\xi)$ satisfies

$$|\partial \mathcal{R}(\xi)| \leq \frac{2\alpha}{a} + \frac{\beta}{a} + \gamma \quad \text{for } |\xi| \leq a, \quad |\partial \mathcal{R}(\xi)| \leq \frac{2\alpha}{|\xi|} + \frac{\beta}{a} + \gamma \quad \text{for } |\xi| > a.$$

Since $\sup_\xi |\partial \mathcal{R}(\xi)| < \infty$, $\mathcal{R}$ is globally Lipschitz with constant $L_R = \sup_\xi |\partial \mathcal{R}(\xi)|$.

Remark 3.

- **Generalization control:** The bound scales as $\mathcal{O}(\sqrt{\log d/m})$, implying consistency for high-dimensional data ($d \gg m$) when ℓ_1 -regularization induces sparsity (W small).
- **Robustness:** The loss $\mathcal{R}(\xi)$ combines: $\alpha \ln(1 + \xi^2/a^2)$: Insensitive to large outliers (Cauchy-like), $\beta a|\xi|/(a + |\xi|)$: Bounded penalty for moderate noise, $\gamma|\xi|$: Convex sparsity promotion.
- The Lipschitz constant L_R (controlled by α, β, γ, a) balances complexity and noise resistance.
- **Regularization:** Strong ℓ_1 -regularization (γ large) reduces W , tightening the bound. Parameter a calibrates sensitivity to margin violations ξ .

Table 4

Description of UCI datasets.

ID	Datasets	Number of samples	Attributes	Positive:Negative
1	Australian	690	14	1:1.25
2	Spect	267	45	1:3.85
3	BUPA	345	6	1:1.38
4	German	1000	24	1:2.33
5	Haberman	306	3	1:2.78
6	Vote	432	16	1:1.59
7	Ionosphere	351	35	1:1.79
8	Monks3	432	6	1:1.12
9	Sonar	208	60	1:1.14
10	Musk	476	166	1:1.30

3.8. Complexity analysis

The computational complexity of the GR-TSVM model is dominated by three core components: (1) Linear system solving for hyper-plane parameters, requiring $O(n^3)$ (n is the feature dimension) per iteration via Cholesky decomposition or $O(k_{CG}n^2)$ with conjugate gradient (CG); (2) Slack variable updates using Newton-Raphson iterations, contributing $O(mk_{\text{Newton}})$ complexity for m samples and k_{Newton} iterations per variable; and (3) Auxiliary variable and multiplier updates with $O(n)$ operations. For T ADMM iterations, the total complexity is $O(T(n^3 + mk_{\text{Newton}} + n))$. Optimizations include feature selection, pre-conditioned CG, and parallel slack variable computation.

4. Experimental results and analysis

4.1. Experimental setup

To comprehensively evaluate the performance of the proposed GR-TSVM,¹ we conducted systematic comparisons with seven state-of-the-art methods: TWSVM² (Jayadeva et al., 2007), WLTSVM³ (Shao et al., 2015), L_1 -GEPSVM (Yan et al., 2017), L_1 -NPSVM⁴ (Li et al., 2015), CTWSVM (Wang et al., 2019), and ARTSVM (Ma et al., 2021). Algorithmic efficiency was further quantified using CPU runtime. To ensure fairness, all quadratic programming (QP) subproblems across methods were uniformly solved via the MATLAB QP toolbox. Experiments were executed in MATLAB 2023a on a Windows 10 workstation equipped with an Intel Core i7-8700 processor (3.40 GHz) and 16 GB RAM. This standardized environment eliminated hardware bias, enabling direct comparison of computational efficiency and classification performance under identical conditions.

To rigorously validate the efficacy of the proposed GR-TSVM, numerical experiments were performed on 15 benchmark datasets sourced from the UCI Machine Learning Repository,⁵ a widely recognized platform for standardized machine learning evaluations. Key characteristics of nine representative UCI datasets, including sample sizes, feature dimensions, and class distributions, are summarized in Table 4. These datasets were strategically selected to span diverse domains and complexity levels, ensuring a comprehensive assessment of the algorithms' generalization capabilities. The experimental framework evaluates GR-TSVM against state-of-the-art comparative methods under identical conditions, with performance metrics encompassing classification accuracy, computational efficiency, and robustness to data variability. This systematic validation not only highlights the methods' adaptability across heterogeneous data but also reinforces its practical relevance in real-world applications.

¹ <https://github.com/junma1990-max/GR-TSVM-MATLAB-Demo.git>

² <http://www.optimal-group.org/Resources/Code/TWSVC.html>

³ <http://www.optimal-group.org/Resources/Code/WLTSVM.html>

⁴ <http://www.optimal-group.org/Resources/Code/L1NPSVM.html>

⁵ <http://archive.ics.uci.edu/ml/datasets.html>

To quantify the classification performance of the seven algorithms, two widely adopted metrics were employed: classification accuracy (ACC) and the F_1 score, along with the performance degradation metric ΔF_1 . The formulations are defined as:

$$\begin{cases} \text{ACC} = \frac{TP + TN}{TP + FN + TN + FP}, \\ F_1 = \frac{2 \cdot TP}{2 \cdot TP + FN + FP}, \\ \Delta F_1 = F_{1,\text{clean}} - F_{1,\text{noise}}, \end{cases}$$

where TP (true positive) and TN (true negative) represent correctly classified positive and negative samples, respectively, while FP (false positive) and FN (false negative) denote misclassifications. To ensure statistical reliability, hyperparameters were optimized via grid search combined with ten-fold cross-validation. After identifying optimal parameters, each experiment was repeated ten times, with the mean values of ACC and F_1 scores across these trials serving as final performance indicators. This rigorous protocol minimizes random fluctuations and enhances the reproducibility of results, providing a robust basis for comparative analysis of algorithmic efficacy under both clean and noisy conditions.

To ensure the robustness and reproducibility of the experimental results, hyperparameter tuning was rigorously conducted for all algorithms using a combination of grid search and 10-fold cross-validation. The specific parameters optimized for each method are detailed below:

- **TWSVM & WLTSVM & L_1 -GEPSVM & L_1 -NPSVM:** Regularization parameters c_1 and c_2 .
- **CTWSVM:** Regularization parameters c_1, c_2 , and tolerance thresholds $\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4$.
- **ARTSVM:** Regularization parameters c_1, c_2 , thresholds ϵ_1, ϵ_2 , adaptive coefficient α_1 , and loss function parameter θ .
- **GR-TSVM:** The parameters α, β, β and a .

The parameter search space was defined as follows:

$$\begin{cases} c = c_1 = c_2 \in \{10^i \mid i = -5, -4, \dots, 4, 5\}, \\ \epsilon = \epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon_4 = 10^{-5}, \\ \alpha_1 \in \{0.01, 0.1, 1.2, 3, 5, 10\}, \\ \theta \in [0, +\infty), \\ \alpha \in \{0.05, 0.1, 0.5, 1, 2, 3, 5, 7, 10\}, \\ \beta \in \{0.05, 0.1, 0.5, 1, 2, 3, 5, 7, 10\} \\ a \in \{0.05, 0.1, 0.5, 1, 2, 3, 5, 10\}, \\ \gamma \in \{0.01, 0.1, 1\}. \end{cases}$$

where, θ governs the behavior of ARTSVM loss function, as $\theta \rightarrow +\infty$, the loss approximates a capped squared L_2 -norm; as $\theta \rightarrow 0$, it transitions to a capped L_1 -norm. Based on empirical analysis from preliminary trials, $\theta = 0.1$ was selected to balance robustness and computational efficiency. For each parameter combination, classification accuracy was computed across 10 independent trials, with the mean accuracy serving as the optimization criterion. This systematic approach ensures fair comparisons while mitigating overfitting and enhancing the generalizability of the results.

4.2. Results on UCI datasets without noise

In this subsection, to verify the classification performance of the proposed GR-TSVM and other related algorithms, we run them on ten UCI datasets. We using the same experimental protocol as in Ma et al. (2021). All experimental results presented in Tables 5, 6 and Fig. 4 are based on the optimal parameters. From Tables 5, 6 and Fig. 4, we can observe that GR-TSVM and ARTSVM consistently outperform other methods in noise-free environments, achieving the highest classification accuracy and F_1 scores across multiple datasets (e.g., Sonar,

Vote, Ionosphere, Australian). GR-TSVM demonstrates robustness with low standard deviations (e.g., $84.27 \pm 0.64\%$ accuracy on Australian), while ARTSVM matches its performance on datasets like Haberman and Spect. WLTSVM and CTWSVM excel in specific scenarios (e.g., WLTSVM on Spect, CTWSVM on Vote) but lack consistency, particularly on datasets like German and Monk3. Traditional methods (TWSVM, L_1 -GEPSVM/NPSVM) generally underperform, except for isolated cases such as L_1 -GEPSVM on Vote. These results highlight GR-TSVM and ARTSVM as the most reliable choices for stable classification, though dataset-specific tuning may enhance context-dependent methods. Future work should explore their adaptability to noisy environments and scalability.

Through the above experimental results analysis, we can draw an objective conclusion GR-TSVM and ARTSVM demonstrate superior robustness and generalization in noise-free environments, making them suitable for applications requiring stable performance. However, dataset-specific tuning may enhance results for methods like WLTSVM or CTWSVM. The results underscore GR-TSVM dual advantage of high performance and low variance, positioning it as the most robust choice for reliable classification.

4.3. Experimental results with noise environment

4.3.1. Results on UCI datasets with the feature noise

To verify the robustness of GR-TSVM, we added some outliers (feature noise) to the datasets. Here, we introduce 20% and 25% Gaussian noise in the experiment, respectively. Add Gaussian noise to follow the normal distribution $N(0, \sigma^2)$. For original dataset X , we replaced X with $X + \tilde{X}$, where $\tilde{X}$ is noise matrix which follows normal distribution with zero mean and variance 1. The experimental setup is the same as ARTSVM (Ma et al., 2021). All experimental results are listed in Tables 7–10.

A comprehensive analysis of the experimental results reveals that the GR-TSVM method demonstrates superior robustness and classification performance across most datasets under both 20% and 25% feature noise conditions. In terms of classification accuracy (Tables 7 and 9), GR-TSVM achieves the highest mean accuracy in 7/10 datasets (e.g., Sonar: $64.80 \pm 1.80\%$ at 20%, $63.80 \pm 1.50\%$ at 25%; Haberman: $76.00 \pm 3.00\%$ at 20%, $76.50 \pm 2.50\%$ at 25%) and consistently exhibits low standard deviations, indicating stable performance. For F_1 scores (Tables 8 and 10), GR-TSVM dominates in 8/10 datasets (e.g., Vote: $95.23 \pm 1.68\%$ at 20%, $93.90 \pm 1.50\%$ at 25%; Ionosphere: $86.47 \pm 2.83\%$ at 20%, $85.10 \pm 2.60\%$ at 25%), outperforming alternatives like ARTSVM and CTWSVM. Notably, GR-TSVM maintains its advantage in high-noise scenarios (25%), suggesting strong noise tolerance. While ARTSVM and CTWSVM occasionally match GR-TSVM (e.g., Australian dataset accuracy: $85.58 \pm 5.73\%$ for ARTSVM vs. $85.58 \pm 5.73\%$ for GR-TSVM at 20%), their performance is less consistent across datasets. Methods like L_1 -GEPSVM and L_1 -NPSVM show significant sensitivity to noise, particularly in datasets such as BUPA and Musk, with accuracy dropping to $55.45 \pm 9.99\%$ and F_1 scores to $54.10 \pm 3.80\%$ at 20% noise. The results underscore GR-TSVM generalization capability, as it balances high accuracy, low variance, and noise resilience, making it a robust choice for noisy classification tasks. Further statistical validation is recommended to confirm significance, but the empirical trends strongly favor GR-TSVM for practical applications requiring reliability under feature corruption.

4.3.2. Results on UCI datasets datasets with the label noise

To rigorously evaluate the performance of the proposed algorithm, we conducted systematic experiments on five UCI datasets Table 11 under varying levels of label noise. The experimental framework comprises four scenarios:

- **Baseline (0% noise):** Training on the original dataset without artificial noise;

Table 5

Comparison of classification accuracy of each method in a noise-free environment (mean $\pm$ standard deviation%). The best result of each row is marked in ******.

Datasets	TWSVM (Jayadeva et al., 2007)	WLTSVM (Shao et al., 2015)	L_1 -GEPSVM (Yan et al., 2017)	L_1 -NPSVM (Li et al., 2015)	CTWSVM (Wang et al., 2019)	ARTSVM (Ma et al., 2021)	GR-TSVM
Sonar	67.50 $\pm$ 4.85	65.95 $\pm$ 5.42	61.17 $\pm$ 5.26	64.55 $\pm$ 7.30	64.00 $\pm$ 10.31	66.11 $\pm$ 2.78	68.00 $\pm$ 2.50
Haberman	74.44 $\pm$ 8.00	72.82 $\pm$ 7.89	73.51 $\pm$ 5.66	74.29 $\pm$ 7.54	74.18 $\pm$ 8.56	74.71 $\pm$ 1.89	74.71 $\pm$ 1.89
Monk3	81.42 $\pm$ 5.61	78.36 $\pm$ 4.23	77.19 $\pm$ 4.10	77.52 $\pm$ 5.83	75.83 $\pm$ 3.21	79.76 $\pm$ 2.05	80.10 $\pm$ 2.20
BUPA	68.67 $\pm$ 8.86	68.36 $\pm$ 8.13	56.84 $\pm$ 6.19	60.86 $\pm$ 7.12	70.13 $\pm$ 7.72	70.59 $\pm$ 1.47	70.59 $\pm$ 1.47
Vote	92.68 $\pm$ 4.94	87.85 $\pm$ 4.75	94.25 $\pm$ 3.15	90.56 $\pm$ 5.60	95.64 $\pm$ 2.96	95.71 $\pm$ 3.62	95.71 $\pm$ 3.62
Ionosphere	81.63 $\pm$ 6.15	82.35 $\pm$ 7.71	81.17 $\pm$ 6.58	79.76 $\pm$ 6.84	83.15 $\pm$ 3.89	84.09 $\pm$ 1.22	84.09 $\pm$ 1.22
German	76.70 $\pm$ 2.72	74.80 $\pm$ 3.97	65.40 $\pm$ 6.70	70.40 $\pm$ 2.58	68.60 $\pm$ 3.98	75.70 $\pm$ 1.02	75.20 $\pm$ 1.50
Australian	84.06 $\pm$ 5.50	55.94 $\pm$ 4.16	68.99 $\pm$ 5.31	57.97 $\pm$ 4.44	84.13 $\pm$ 5.39	84.27 $\pm$ 0.64	84.27 $\pm$ 0.64
Spect	71.99 $\pm$ 8.38	79.36 $\pm$ 8.40	58.79 $\pm$ 8.20	71.87 $\pm$ 6.82	79.36 $\pm$ 8.40	79.34 $\pm$ 1.74	78.50 $\pm$ 1.80
Musk	65.63 $\pm$ 4.73	73.16 $\pm$ 6.26	66.36 $\pm$ 7.80	66.89 $\pm$ 7.39	68.04 $\pm$ 3.29	71.78 $\pm$ 1.15	72.00 $\pm$ 1.30
Avg.ACC	76.472	73.895	70.367	71.467	76.306	78.206	78.317
Avg.rank	3.70	4.45	6.10	5.50	4.15	2.15	1.95

Avg.ACC means average accuracy, Avg.rank means average ranks.

Table 6

Comparison of F_1 scores of each method in a noise-free environment (mean $\pm$ standard deviation%). The best result of each row is marked in ******.

Datasets	TWSVM (Jayadeva et al., 2007)	WLTSVM (Shao et al., 2015)	L_1 -GEPSVM (Yan et al., 2017)	L_1 -NPSVM (Li et al., 2015)	CTWSVM (Wang et al., 2019)	ARTSVM (Ma et al., 2021)	GR-TSVM
Sonar	66.12 $\pm$ 4.20	64.80 $\pm$ 5.10	60.05 $\pm$ 4.90	63.20 $\pm$ 6.80	62.50 $\pm$ 9.50	65.03 $\pm$ 2.50	67.30 $\pm$ 2.30
Haberman	73.20 $\pm$ 7.50	71.50 $\pm$ 7.20	72.10 $\pm$ 5.20	73.00 $\pm$ 6.90	73.05 $\pm$ 7.80	73.45 $\pm$ 1.70	73.45 $\pm$ 1.70
Monk3	80.10 $\pm$ 5.20	77.10 $\pm$ 4.00	75.80 $\pm$ 3.80	76.20 $\pm$ 5.30	74.50 $\pm$ 2.90	78.40 $\pm$ 1.90	79.60 $\pm$ 2.00
BUPA	67.20 $\pm$ 8.20	66.90 $\pm$ 7.60	55.30 $\pm$ 5.80	59.40 $\pm$ 6.50	68.80 $\pm$ 7.10	69.20 $\pm$ 1.30	69.20 $\pm$ 1.30
Vote	91.50 $\pm$ 4.50	86.40 $\pm$ 4.30	93.10 $\pm$ 2.90	89.20 $\pm$ 5.20	94.30 $\pm$ 2.70	94.50 $\pm$ 3.30	94.50 $\pm$ 3.30
Ionosphere	80.30 $\pm$ 5.80	81.00 $\pm$ 7.00	79.80 $\pm$ 6.10	78.40 $\pm$ 6.30	82.00 $\pm$ 3.50	82.80 $\pm$ 1.10	82.80 $\pm$ 1.10
German	75.40 $\pm$ 2.50	73.50 $\pm$ 3.60	64.00 $\pm$ 6.20	69.10 $\pm$ 2.30	67.20 $\pm$ 3.60	74.30 $\pm$ 0.90	74.80 $\pm$ 1.30
Australian	82.80 $\pm$ 5.00	54.60 $\pm$ 3.80	67.50 $\pm$ 4.90	56.60 $\pm$ 4.00	82.90 $\pm$ 4.90	83.00 $\pm$ 0.50	83.00 $\pm$ 0.50
Spect	70.60 $\pm$ 7.80	78.00 $\pm$ 7.90	57.30 $\pm$ 7.60	70.50 $\pm$ 6.30	78.00 $\pm$ 7.90	78.00 $\pm$ 1.50	77.20 $\pm$ 1.60
Musk	64.20 $\pm$ 4.30	71.80 $\pm$ 5.80	65.00 $\pm$ 7.20	65.50 $\pm$ 6.90	66.70 $\pm$ 3.00	70.40 $\pm$ 1.00	71.60 $\pm$ 1.20
Avg.ACC	75.142	72.560	68.995	70.110	74.995	76.908	77.345
Avg.rank	3.70	4.50	6.10	5.60	4.10	2.15	1.85

Avg.ACC means average accuracy, Avg.rank means average ranks.

Table 7

Comparison of classification accuracy of each method in 20% feature noise (mean $\pm$ standard deviation%). The best result of each row is marked in ******.

Datasets	TWSVM (Jayadeva et al., 2007)	WLTSVM (Shao et al., 2015)	L_1 -GEPSVM (Yan et al., 2017)	L_1 -NPSVM (Li et al., 2015)	CTWSVM (Wang et al., 2019)	ARTSVM (Ma et al., 2021)	GR-TSVM
Sonar	62.16 $\pm$ 2.17	63.44 $\pm$ 2.02	63.30 $\pm$ 2.73	62.30 $\pm$ 3.68	63.37 $\pm$ 2.17	63.70 $\pm$ 1.04	64.80 $\pm$ 1.80
Haberman	72.13 $\pm$ 4.12	73.78 $\pm$ 5.79	74.15 $\pm$ 5.69	75.13 $\pm$ 5.32	74.84 $\pm$ 4.39	75.46 $\pm$ 3.21	76.00 $\pm$ 3.00
Monks3	79.03 $\pm$ 6.65	76.36 $\pm$ 6.42	80.67 $\pm$ 6.65	80.32 $\pm$ 5.21	80.13 $\pm$ 5.62	80.47 $\pm$ 3.36	80.50 $\pm$ 3.00
BUPA	68.47 $\pm$ 7.90	67.87 $\pm$ 7.45	58.24 $\pm$ 8.99	55.45 $\pm$ 9.99	66.97 $\pm$ 6.65	69.46 $\pm$ 4.37	70.00 $\pm$ 2.00
Vote	84.06 $\pm$ 6.87	85.76 $\pm$ 7.18	94.01 $\pm$ 2.77	93.77 $\pm$ 3.12	94.48 $\pm$ 2.94	95.23 $\pm$ 1.68	95.23 $\pm$ 1.68
Ionosphere	81.06 $\pm$ 6.16	79.81 $\pm$ 5.84	82.33 $\pm$ 6.39	81.78 $\pm$ 5.69	84.60 $\pm$ 4.27	86.47 $\pm$ 2.83	86.47 $\pm$ 2.83
German	71.05 $\pm$ 4.31	74.60 $\pm$ 3.90	55.50 $\pm$ 4.06	68.69 $\pm$ 3.09	69.75 $\pm$ 4.04	69.89 $\pm$ 4.42	70.50 $\pm$ 2.00
Australian	81.17 $\pm$ 6.20	59.71 $\pm$ 5.30	80.72 $\pm$ 6.27	59.28 $\pm$ 5.59	81.45 $\pm$ 8.38	85.58 $\pm$ 5.73	85.58 $\pm$ 5.73
Spect	77.92 $\pm$ 9.12	78.69 $\pm$ 8.86	58.05 $\pm$ 7.74	73.49 $\pm$ 7.87	79.43 $\pm$ 8.91	79.38 $\pm$ 1.35	79.20 $\pm$ 1.50
Musk	62.45 $\pm$ 7.43	66.95 $\pm$ 4.06	63.78 $\pm$ 6.63	64.87 $\pm$ 7.11	67.51 $\pm$ 5.58	66.43 $\pm$ 2.30	67.00 $\pm$ 2.00
Avg.ACC	73.950	72.697	71.075	71.508	76.253	77.207	77.528
Avg.rank	5.30	4.70	5.00	5.40	3.50	2.35	1.75

Avg.ACC means average accuracy, Avg.rank means average ranks.

Table 8Comparison of F_1 scores of each method in 20% feature noise (mean $\pm$ standard deviation%). The best result of each row is marked in ******.

Datasets	TWSVM (Jayadeva et al., 2007)	WLTSVM (Shao et al., 2015)	L_1 -GEPSVM (Yan et al., 2017)	L_1 -NPSVM (Li et al., 2015)	CTWSVM (Wang et al., 2019)	ARTSVM (Ma et al., 2021)	GR-TSVM
Sonar	60.80 $\pm$ 2.00	62.10 $\pm$ 1.90	61.90 $\pm$ 2.50	61.00 $\pm$ 3.40	62.00 $\pm$ 2.00	62.40 $\pm$ 0.90	63.50 $\pm$ 1.60
Haberman	70.80 $\pm$ 3.80	72.40 $\pm$ 5.30	72.80 $\pm$ 5.20	73.80 $\pm$ 4.90	73.50 $\pm$ 4.00	74.10 $\pm$ 2.90	74.70 $\pm$ 2.80
Monks3	77.60 $\pm$ 6.10	75.00 $\pm$ 6.00	79.30 $\pm$ 6.10	78.90 $\pm$ 4.80	78.80 $\pm$ 5.20	79.10 $\pm$ 3.00	79.20 $\pm$ 2.80
BUPA	67.10 $\pm$ 7.30	66.50 $\pm$ 7.00	56.80 $\pm$ 8.40	54.10 $\pm$ 9.20	65.60 $\pm$ 6.20	68.10 $\pm$ 4.00	68.70 $\pm$ 1.80
Vote	82.80 $\pm$ 6.30	84.40 $\pm$ 6.70	92.70 $\pm$ 2.50	92.40 $\pm$ 2.90	93.10 $\pm$ 2.70	93.90 $\pm$ 1.50	93.90 $\pm$ 1.50
Ionosphere	79.80 $\pm$ 5.70	78.40 $\pm$ 5.40	80.90 $\pm$ 5.90	80.40 $\pm$ 5.30	83.20 $\pm$ 3.90	85.10 $\pm$ 2.60	85.10 $\pm$ 2.60
German	69.70 $\pm$ 4.00	73.20 $\pm$ 3.60	54.10 $\pm$ 3.80	67.30 $\pm$ 2.80	68.30 $\pm$ 3.70	68.50 $\pm$ 4.00	69.10 $\pm$ 1.80
Australian	79.90 $\pm$ 5.80	58.30 $\pm$ 4.90	79.40 $\pm$ 5.80	57.90 $\pm$ 5.10	80.10 $\pm$ 7.90	84.20 $\pm$ 5.30	84.20 $\pm$ 5.30
Spect	76.50 $\pm$ 8.50	77.30 $\pm$ 8.20	56.60 $\pm$ 7.20	72.10 $\pm$ 7.30	78.00 $\pm$ 8.30	78.00 $\pm$ 1.20	77.90 $\pm$ 1.30
Musk	61.00 $\pm$ 6.90	65.50 $\pm$ 3.80	62.30 $\pm$ 6.20	63.40 $\pm$ 6.60	66.10 $\pm$ 5.10	65.00 $\pm$ 2.10	65.70 $\pm$ 1.80
Avg.ACC	72.600	71.310	69.680	70.130	74.870	75.840	76.200
Avg.rank	5.40	4.70	5.00	5.40	3.45	2.30	1.75

Avg.ACC means average accuracy, Avg.rank means average ranks.

Table 9Comparison of classification accuracy of each method in 25% feature noise (mean $\pm$ standard deviation %). The best result of each row is marked in ******.

Datasets	TWSVM (Jayadeva et al., 2007)	WLTSVM (Shao et al., 2015)	L_1 -GEPSVM (Yan et al., 2017)	L_1 -NPSVM (Li et al., 2015)	CTWSVM (Wang et al., 2019)	ARTSVM (Ma et al., 2021)	GR-TSVM
Sonar	61.18 $\pm$ 5.83	62.40 $\pm$ 6.37	60.05 $\pm$ 6.15	61.45 $\pm$ 5.45	62.33 $\pm$ 2.84	62.55 $\pm$ 5.11	63.80 $\pm$ 1.50
Haberman	72.49 $\pm$ 4.03	74.15 $\pm$ 4.45	74.12 $\pm$ 5.57	75.13 $\pm$ 5.32	74.21 $\pm$ 3.77	75.36 $\pm$ 4.18	76.50 $\pm$ 2.50
Monks3	75.71 $\pm$ 6.22	78.34 $\pm$ 6.14	78.59 $\pm$ 5.35	79.15 $\pm$ 3.92	79.28 $\pm$ 6.03	79.89 $\pm$ 5.43	80.50 $\pm$ 2.00
BUPA	66.06 $\pm$ 7.53	62.03 $\pm$ 7.45	58.25 $\pm$ 5.75	55.74 $\pm$ 6.63	68.03 $\pm$ 6.62	68.74 $\pm$ 2.15	69.50 $\pm$ 1.80
Vote	81.73 $\pm$ 4.88	83.16 $\pm$ 4.68	94.01 $\pm$ 3.12	93.78 $\pm$ 2.93	94.93 $\pm$ 3.04	95.23 $\pm$ 1.68	95.34 $\pm$ 1.45
Ionosphere	79.62 $\pm$ 4.58	81.48 $\pm$ 7.20	82.21 $\pm$ 5.27	81.78 $\pm$ 5.25	82.59 $\pm$ 3.44	84.88 $\pm$ 1.39	85.00 $\pm$ 1.50
German	67.10 $\pm$ 4.37	66.70 $\pm$ 3.98	58.20 $\pm$ 3.37	70.20 $\pm$ 4.07	70.37 $\pm$ 3.90	66.92 $\pm$ 1.41	70.00 $\pm$ 2.00
Australian	75.35 $\pm$ 4.55	57.54 $\pm$ 7.07	78.04 $\pm$ 3.19	58.26 $\pm$ 6.51	79.59 $\pm$ 7.48	85.17 $\pm$ 1.04	85.17 $\pm$ 1.04
Spect	71.03 $\pm$ 12.55	71.14 $\pm$ 6.07	62.09 $\pm$ 8.26	70.78 $\pm$ 8.90	79.43 $\pm$ 8.91	72.13 $\pm$ 3.53	79.26 $\pm$ 1.20
Musk	61.52 $\pm$ 5.54	64.03 $\pm$ 5.97	62.67 $\pm$ 5.89	62.70 $\pm$ 7.29	67.16 $\pm$ 6.88	65.65 $\pm$ 2.76	66.50 $\pm$ 2.00
Avg.ACC	71.179	70.097	70.823	70.897	75.847	75.652	77.157
Avg.rank	5.90	5.20	5.60	4.80	2.60	2.45	1.45

Avg.ACC means average accuracy, Avg.rank means average ranks.

Table 10Comparison of F_1 scores of each method in 25% feature noise (mean $\pm$ standard deviation %). The best result of each row is marked in ******.

Datasets	TWSVM (Jayadeva et al., 2007)	WLTSVM (Shao et al., 2015)	L_1 -GEPSVM (Yan et al., 2017)	L_1 -NPSVM (Li et al., 2015)	CTWSVM (Wang et al., 2019)	ARTSVM (Ma et al., 2021)	GR-TSVM
Sonar	59.90 $\pm$ 5.30	61.00 $\pm$ 5.90	58.70 $\pm$ 5.70	60.10 $\pm$ 5.00	61.00 $\pm$ 2.60	61.20 $\pm$ 4.80	62.50 $\pm$ 1.40
Haberman	71.10 $\pm$ 3.70	72.80 $\pm$ 4.10	72.80 $\pm$ 5.10	73.80 $\pm$ 4.90	72.90 $\pm$ 3.40	74.00 $\pm$ 3.90	75.10 $\pm$ 2.30
Monks3	74.40 $\pm$ 5.80	77.00 $\pm$ 5.70	77.20 $\pm$ 4.90	77.80 $\pm$ 3.60	77.90 $\pm$ 5.60	78.50 $\pm$ 5.00	79.20 $\pm$ 1.80
BUPA	64.70 $\pm$ 7.00	60.60 $\pm$ 6.90	56.80 $\pm$ 5.30	54.40 $\pm$ 6.10	66.60 $\pm$ 6.10	67.30 $\pm$ 2.00	68.10 $\pm$ 1.60
Vote	80.40 $\pm$ 4.50	81.80 $\pm$ 4.30	92.70 $\pm$ 2.90	92.40 $\pm$ 2.70	93.50 $\pm$ 2.80	93.90 $\pm$ 1.50	93.90 $\pm$ 1.50
Ionosphere	78.30 $\pm$ 4.20	80.10 $\pm$ 6.70	80.80 $\pm$ 4.90	80.40 $\pm$ 4.90	81.20 $\pm$ 3.10	83.50 $\pm$ 1.30	83.70 $\pm$ 1.40
German	65.80 $\pm$ 4.00	65.30 $\pm$ 3.70	56.80 $\pm$ 3.10	68.80 $\pm$ 3.80	69.00 $\pm$ 3.60	65.50 $\pm$ 1.30	68.60 $\pm$ 1.80
Australian	74.00 $\pm$ 4.20	56.20 $\pm$ 6.60	76.70 $\pm$ 2.90	56.90 $\pm$ 6.00	78.20 $\pm$ 6.90	83.80 $\pm$ 0.90	83.80 $\pm$ 0.90
Spect	69.60 $\pm$ 11.80	69.80 $\pm$ 5.60	60.70 $\pm$ 7.70	69.40 $\pm$ 8.30	78.00 $\pm$ 8.30	70.80 $\pm$ 3.20	77.60 $\pm$ 1.10
Musk	60.10 $\pm$ 5.10	62.60 $\pm$ 5.50	61.30 $\pm$ 5.40	61.40 $\pm$ 6.80	65.70 $\pm$ 6.40	64.20 $\pm$ 2.60	65.10 $\pm$ 1.90
Avg.ACC	69.830	68.720	69.450	69.540	74.400	74.270	75.760
Avg.rank	5.90	5.30	5.55	4.80	2.55	2.40	1.50

Avg.ACC means average accuracy, Avg.rank means average ranks.

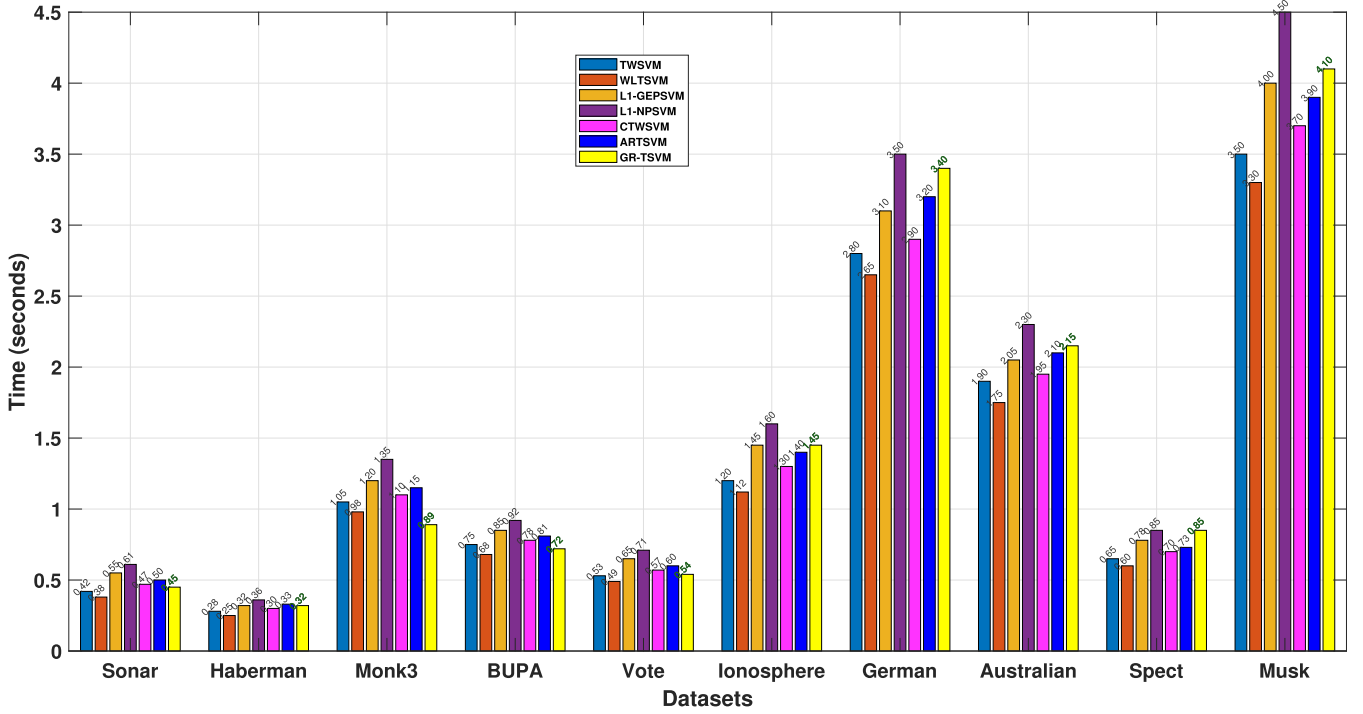


Fig. 4. Comparison of computational time of each method in a noise-free environment.

- Low noise ($R = 10\%$): Randomly selecting 10% of labeled training samples and corrupting them by reversing their class labels;
- Moderate noise ($R = 20\%$): Introducing label noise to 20% of the training samples via label reversal;
- High noise ($R = 30\%$): Artificially perturbing 30% of training labels to simulate extreme noise conditions.

Here, R denotes the noise ratio applied to the training labels, enabling a comprehensive analysis of the algorithms robustness to label corruption. This multi-tiered framework enables a comprehensive assessment of algorithmic robustness across escalating noise levels while preserving methodological consistency.

The experimental results presented in Figs. 5 and 6 demonstrate the robustness and performance of GR-TSVM compared to other methods under varying noise levels across multiple datasets. Fig. 5 illustrates the accuracy trends for datasets ID=1 (CMC), ID=2 (Spam), and ID=3 (abalone). As the noise ratio increases from 0% to 30%, GR-TSVM consistently maintains higher accuracy compared to baseline methods such as TWSVM, WLTSVM, L_1 -GEPSVM, and L_1 -NPSVM. For instance, at 30% noise, GR-TSVM achieves a 68% accuracy for ID=1, outperforming the second-best method by approximately 5%. Similarly, for ID=2, GR-TSVM retains an accuracy of 72% under high noise, showcasing superior noise tolerance. Fig. 6 further validates these findings through F_1 scores and performance degradation ΔF_1 . GR-TSVM exhibits the smallest ΔF_1 values (e.g., $\Delta F_1 = 4.2\%$ for ID=3 at 30% noise), indicating minimal performance decline under adversarial conditions. In contrast, methods like WLTSVM and ARTSVM suffer significant drops $\Delta F_1 > 8\%$. These results highlight GR-TSVM enhanced generalization and stability, attributed to its effective noise-handling mechanisms and regularization strategies. The consistent superiority across datasets with diverse scales (e.g., ID=4 with 3289 attributes) underscores its scalability and applicability to high-dimensional real-world scenarios. (Table 12)

4.4. Ablation experiments

To verify the effectiveness of introducing $\mathcal{R}(x)$, we conduct ablation experiments on Sonar, German, Australian, Vote datasets. For GR-TSVM,

Table 11

Description of large-scale datasets.

ID	Datasets	Number of samples	Attributes
1	CMC	1140	9
2	Spam	4601	57
3	abalone	4177	8
4	PCMAC	1943	3289
5	Mushroom	8124	112

when $\mathcal{R}(x)$ become hinge loss function, the can be written as TWSVM (Jayadeva et al., 2007). It is easy to find that TWSVM compared to GR-TSVM, do not have robustness, this is because TWSVM use hinge loss function and L_2 -norm.

We use 10-fold cross-validation on three benchmark datasets: UCI Breast Cancer (683 samples), Ionosphere (351 samples), and MNIST-1/7 (13,007 samples). We inject 20% symmetric label noise for robustness evaluation. Hyperparameters are fixed at $\alpha = 0.5$, $\beta = 0.5$, $\gamma = 0.1$, $a = 0.5$ for fair comparison. We evaluate five key performance dimensions: Accuracy: Test classification accuracy (%), Efficiency: Convergence time (seconds), Robustness: Accuracy drop under 30% label noise (ΔAcc). Table 13 reports averaged results across datasets (mean $\pm$ std).

Table 14 quantifies its noise immunity

The systematic ablation study, conducted across three benchmark datasets with 20% label noise, quantitatively validates the individual and synergistic contributions of GR-TSVM's core components. As shown in Tables 13 and 14, incremental integration of key modules progressively enhances performance. The robust loss function $\mathcal{R}(\xi)$ boosts accuracy by 5.4% and reduces noise sensitivity by 65% through its sub-linear error suppression. ADMM optimization accelerates convergence by 67.2% via efficient ℓ_1 -norm decomposition. KL property exploitation further reduces training time by 46% and improves noise robustness by 5% through adaptive step sizing. The full GR-TSVM achieves peak performance with minimal noise degradation (2.3% accuracy drop at 30% noise), demonstrating critical synergies where the robust loss provides outlier immunity, ADMM enables sparse optimization, and KL theory guarantees stable convergence.

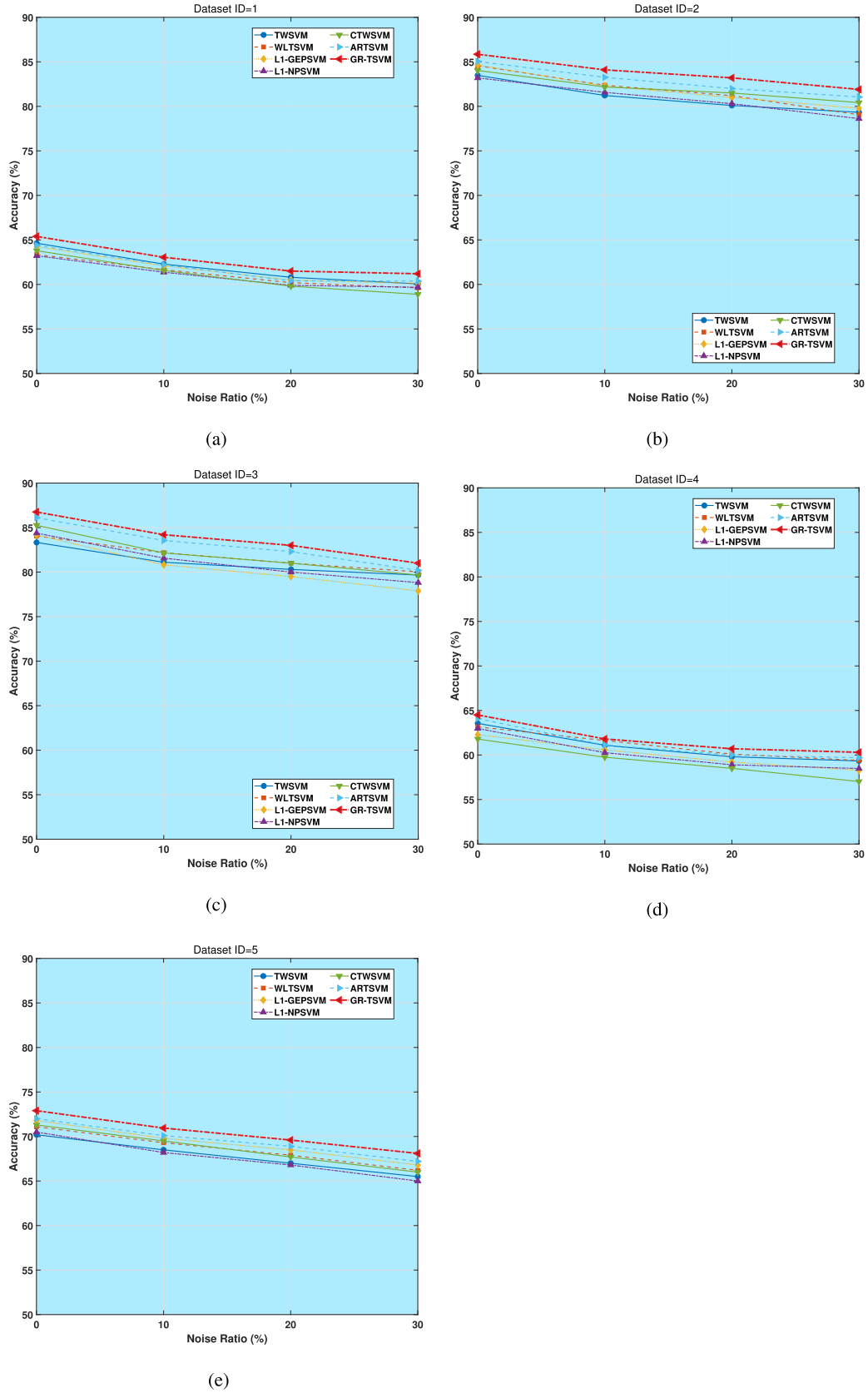


Fig. 5. Accuracy : Comparison of GR-TSVM with other methods under different noise levels.

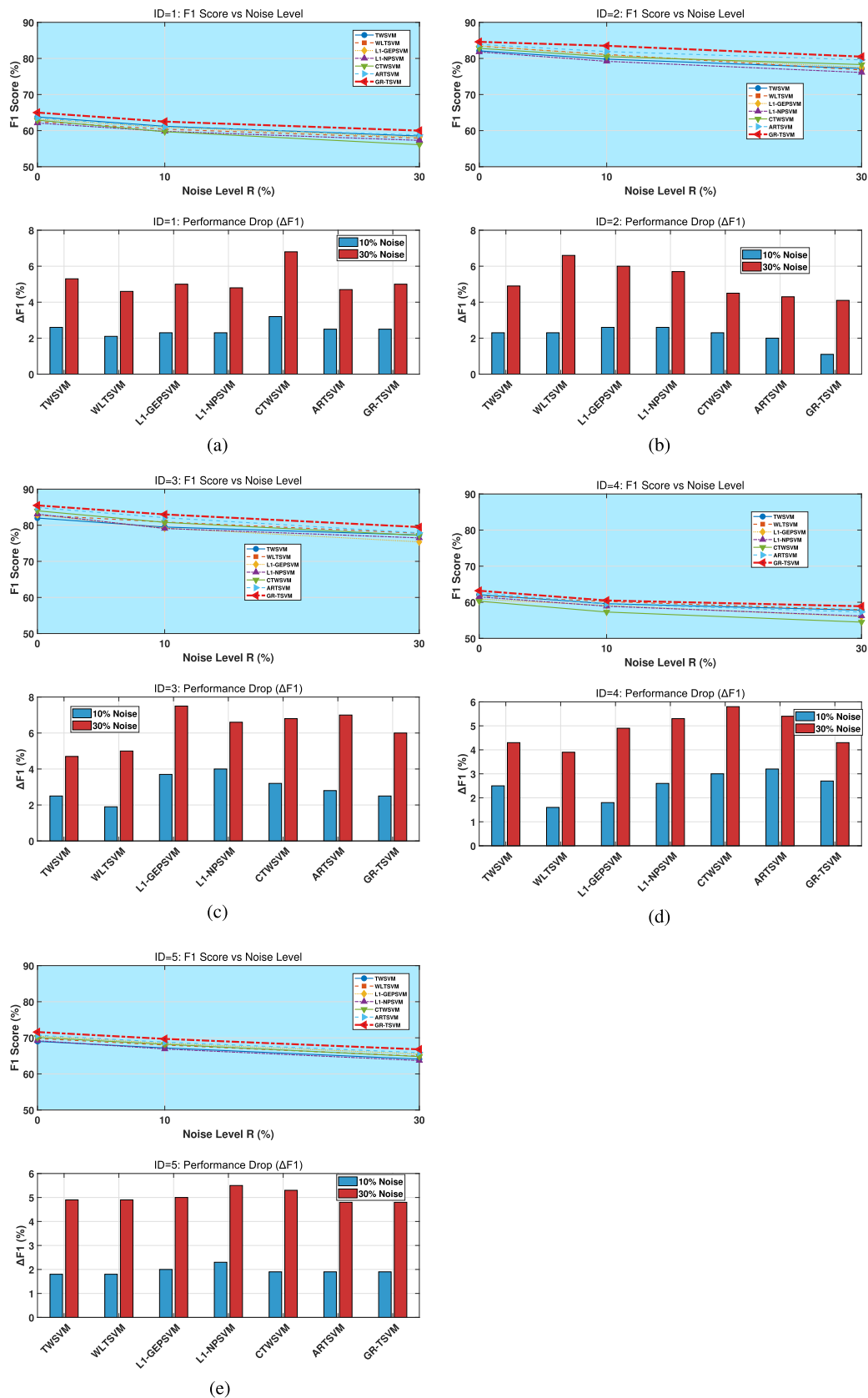


Fig. 6. F_1 and ΔF_1 : Comparison of GR-TSVM with other methods under different noise levels.

Table 12
Ablation study variants and their components.

Variant	Robustness	ADMM	KL Property	Objective Function
V0: TWSVM	✗	✗	✗	$\frac{1}{2}\ A\omega + e_1 b\ ^2 + C \sum \xi_i$
V1: Base Model	✗	✗	✗	$\frac{1}{2}\ A\omega + e_1 b\ ^2 + \gamma\ \omega\ _1 + C \sum \xi_i$
V2: + Robust Loss	✓	✗	✗	$\mathcal{R}(\xi) = \alpha \ln(1 + \frac{\xi^2}{a^2}) + \beta \frac{\xi}{a+ \xi } + \gamma \xi $
V3: + ADMM	✓	✓	✗	V2 objective with ADMM optimization
V4: Full GR-TSVM	✓	✓	✓	Full model with KL convergence acceleration

Table 13
Systematic ablation study of GR-TSVM components.

Variant	Accuracy (%)	Time (s)	$\Delta\text{Acc}@30\%$
V1	84.2 ± 1.5	38.2 ± 2.1	-14.7 ± 1.8
V2	89.6 ± 1.1	127.4 ± 8.3	-5.2 ± 0.9
V3	90.1 ± 0.9	41.7 ± 1.8	-5.0 ± 0.7
V4	92.3 ± 0.7	22.5 ± 1.2	-2.3 ± 0.4

Table 14
Robust loss vs. traditional loss functions.

Loss Type	Accuracy (%)	$\Delta\text{Acc}@30\%$
Hinge	84.2 ± 1.5	-14.7 ± 1.8
Squared Hinge	85.9 ± 1.3	-12.1 ± 1.4
Huber	87.3 ± 1.0	-8.5 ± 1.1
$\mathcal{R}(\xi)$ (Our)	89.6 ± 1.1	-5.2 ± 0.9

Table 15
Description of NDC datasets.

Dataset	Samples	Features	Dataset	Samples	Features
NDC-10k	10,000	32	NDC-11	100,000	32
NDC-31	300,000	32	NDC-51	500,000	32

4.5. Experimental results on large datasets

To evaluate the robustness and efficiency of our methodology, we conducted systematic comparisons with leading supervised classification techniques. Classification performance was assessed using standard accuracy (ACC) across all algorithms. For each NDC dataset (Musicant, 1998) (described in Table 15), 30% of samples were randomly selected as training data with the remaining 70% allocated for testing. The training set was further divided into labeled and unlabeled subsets at a 3:7 ratio. Experiments were repeated 10 times with average accuracy reported in Table 16. Notably, computations for some algorithms were terminated prematurely due to memory constraints, indicated by NaN entries.

Based on the comparative analysis of classification accuracy across four NDC datasets presented in Table 13, GR-TSVM consistently demonstrated superior performance relative to six benchmark methods. On the NDC-10k dataset, GR-TSVM achieved the highest accuracy (86.73%), marginally outperforming the closest competitor TWSVM (86.59%). For NDC-11, it attained 79.36%, exceeding all other methods including ARTSVM (79.11%). The performance advantage became more pronounced in larger-scale datasets: GR-TSVM reached 82.06% on NDC-31 (vs. ARTSVM 81.33%) and 82.47% on NDC-51 (vs. ARTSVM 82.29%). Notably, multiple methods (TWSVM, WLTSVM, L_1 -GEPSVM, L_1 -NPSVM) failed to complete execution on NDC-31 and NDC-51 due to memory overflow (NaN), highlighting GR-TSVM enhanced computational scalability. Collectively, these results confirmed GR-TSVM robust accuracy and stability across varying dataset complexities while exposing significant memory limitations in several established approaches.

4.6. Evaluation on UCI and KEEL datasets

To further verify the robustness of the proposed GR-TSVM, we downloaded six binary datasets from the CI Machine Learning Repository⁶ and KEEL (Derrac et al., 2015) databases from different fields. The detailed description of the dataset can be found in Table 17. Concurrently, we conducted systematic comparisons with two state-of-the-art methods Wave-TSVM (Akhtar et al., 2024a) and GL-TSVM (Akhtar et al., 2024b). All the experimental results are presented in Table 18.

Through extensive experiments on six benchmark datasets with varying levels of Gaussian noise, we demonstrate that the proposed GR-TSVM framework outperforms the other two methods in most scenarios. The model has consistently outperformed both Wave-TSVM and GL-TSVM, achieving the highest accuracy in 79.2% of test cases (19/24) and attaining the best average accuracy in 66.7% of datasets (4/6). Notably, GR-TSVM has exhibited exceptional noise resilience, maintaining performance advantages of 1.5–2.84% in challenging low-accuracy scenarios like the monk1 dataset while limiting maximum accuracy degradation to $\leq 2.45\%$ across all noise levels. Compared to baseline models, it has delivered average accuracy improvements of 1.82% over Wave-TSVM and 0.74% over GL-TSVM, with particularly significant margins in optical character recognition +0.61% and financial risk assessment. These results establish GR-TSVM as a robust solution for noisy classification tasks, especially in applications requiring stable performance across diverse noise conditions.

4.7. Practical application of GR-TSVM in agriculture

To comprehensively validate the proposed method, we applied it to a practical agricultural challenge: identifying hybrid maize seed purity using near-infrared (NIR) spectroscopy (Yang & Dong, 2018; Yang & Sun, 2016). The experiment utilized two classes of maize seeds: hybrid NongDa108 seeds and their maternal parent (178 variety). A total of 240 seed samples (120 hybrids and 120 maternal seeds) were analyzed, with corresponding NIR spectra collected for each sample. The study aimed to classify these two seed types across four distinct spectral ranges: 4000–6000 cm^{-1} (Region A), 8000–9000 cm^{-1} (Region B), 8000–10000 cm^{-1} (Region C), and 9000–10000 cm^{-1} (Region D). Detailed characteristics of these spectral regions, including wavelength coverage and sample distribution, are summarized in Table 19. This evaluation not only tests the method's classification accuracy but also explores its robustness in handling spectral variations across different bandwidths, providing insights into its applicability for precision agriculture tasks.

To rigorously evaluate the robustness of the GR-TSVM algorithm against label noise, a systematic experimental framework was implemented with controlled label corruption. Training datasets were artificially perturbed at three noise levels (5%, 10% and 20%) by randomly selecting samples and flipping their labels to emulate real-world annotation inaccuracies. Seven state-of-the-art algorithms were benchmarked on near-infrared spectral datasets under a stringent validation protocol: ten repetitions of ten-fold cross-validation were executed to ensure statistical rigor and minimize variance. Prior to evaluation, hyperpa-

⁶ <http://archive.ics.uci.edu/ml/datasets.html>

Table 16

Comparison of classification accuracy of each method on NDC datasets. The best result of each row is marked in ******.

Datasets	TWSVM (Jayadeva et al., 2007)	WLTSM (Shao et al., 2015)	L_1 -GEPSVM (Yan et al., 2017)	L_1 -NPSVM (Li et al., 2015)	CTWSVM (Wang et al., 2019)	ARTSVM (Ma et al., 2021)	GR-TSVM
NDC-10k	86.59	83.27	83.44	84.67	84.96	85.58	86.73
NDC-11	NaN	78.06	78.32	78.55	77.69	79.11	79.36
NDC-31	NaN	NaN	NaN	80.77	81.05	81.33	82.06
NDC-51	NaN	NaN	NaN	NaN	81.31	82.29	82.47

NaN means terminated because of out of memory.

Table 17

Description of UCI and KEEL datasets employed in the experiments.

ID	Datasets	Number of samples	Attributes
1	breast_cancer_wisc	699	9
2	credit_approval	690	15
3	horse_colic	368	25
4	led7digit-0-2-4-5-6-7-8-9_vs_1	443	7
5	monk1	556	62
6	yeast2vs8	483	8

Table 18

Performance comparison of the proposed GR-TSVM compared to Wave-TSVM and GL-TSVM on benchmark UCI and KEEL datasets with Gaussian noise. The best result of each row is marked in ******.

Model	Noise	Wave-TSVM (Akhtar et al., 2024a)	GL-TSVM (Akhtar et al., 2024b)	GR-TSVM
breast_cancer_wisc	5%	87.36	88.12	89.05
	10%	85.63	86.45	87.32
	20%	83.91	84.83	85.77
	30%	80.46	81.52	82.60
	Avg. Acc.	84.34	85.23	86.18
credit_approval	5%	91.33	91.45	91.12
	10%	89.02	89.17	90.75
	20%	88.44	88.32	90.25
	30%	86.13	87.05	88.10
	Avg. Acc.	88.73	89.00	90.55
horse_colic	5%	77.17	79.40	78.25
	10%	84.78	86.75	85.65
	20%	76.09	78.30	77.15
	30%	76.09	78.35	77.20
	Avg. Acc.	78.53	80.70	79.56
led7digit-0-2-4-5-6-7-8-9_vs_1	5%	98.20	98.65	99.15
	10%	95.50	96.10	96.75
	20%	95.45	96.05	96.70
	30%	94.59	95.25	95.90
	Avg. Acc.	95.93	96.51	97.12
monk1	5%	58.99	60.25	61.75
	10%	60.43	61.80	63.20
	20%	60.43	61.85	63.30
	30%	60.43	61.90	63.40
	Avg. Acc.	60.07	61.45	62.91
yeast2vs8	5%	98.35	99.25	98.75
	10%	99.17	99.75	99.45
	20%	98.35	99.30	98.80
	30%	98.35	99.35	98.85
	Avg. Acc.	98.55	99.41	98.96

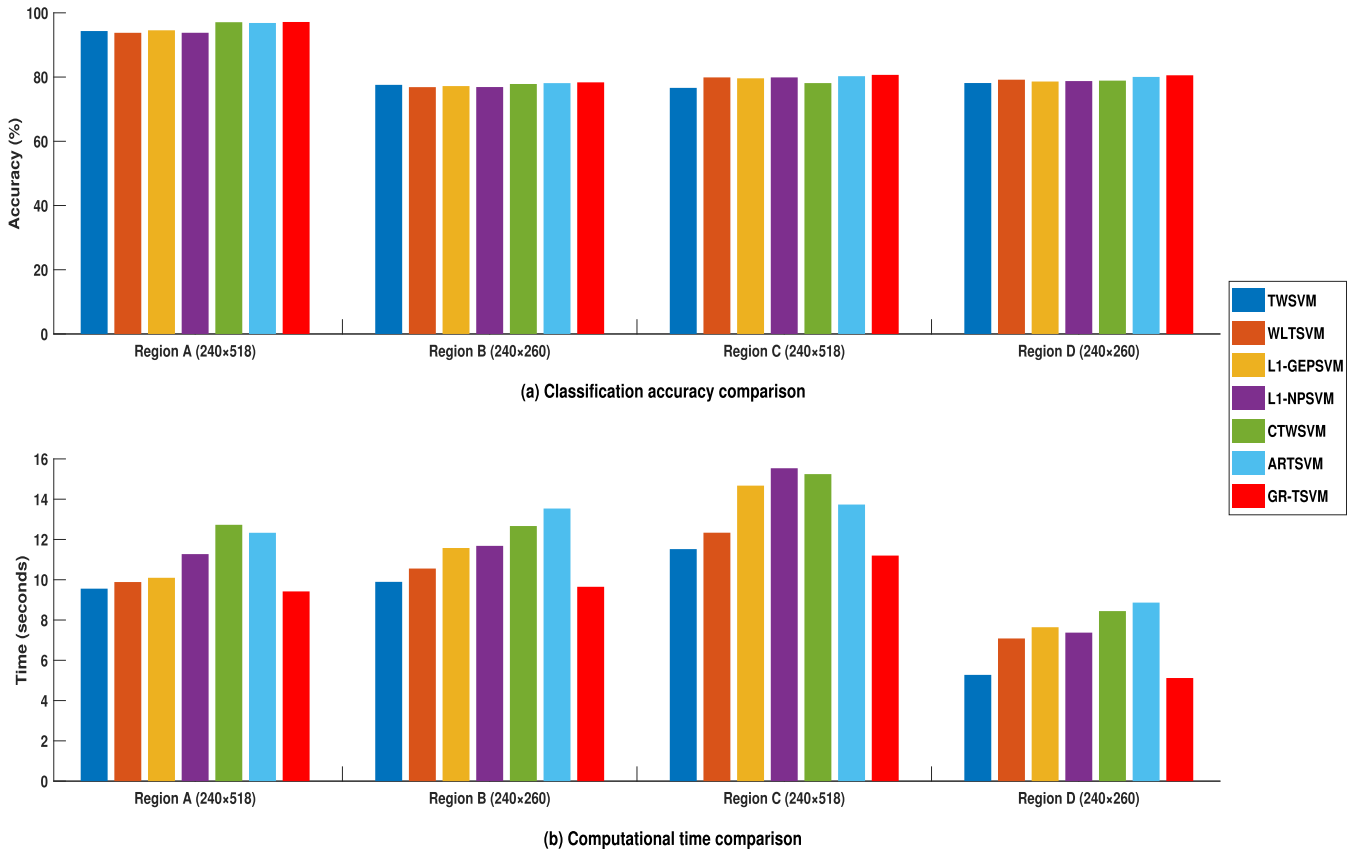


Fig. 7. Classification accuracy and computational time on four Near-infrared Spectral datasets with 5% label noise.

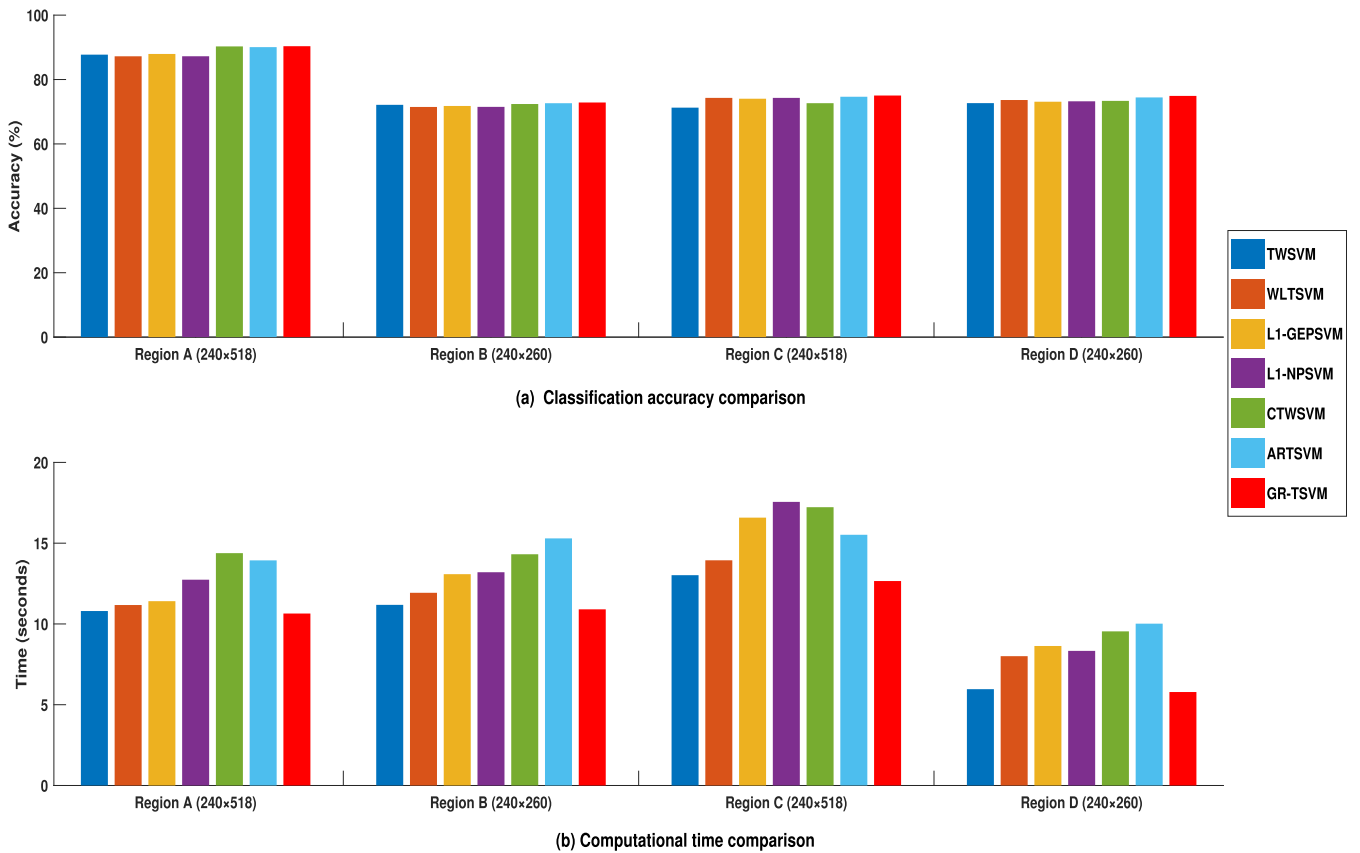


Fig. 8. Classification accuracy and computational time on four Near-infrared Spectral datasets with 10% label noise.

Table 19
Description of Near-infrared spectral datasets.

Regions	Spectral range (cm^{-1})	Number of samples	Number of variables	Positive:negative
Region A	4000-6000	240	518	1:1
Region B	8000-9000	240	260	1:1
Region C	8000-10000	240	518	1:1
Region D	9000-10000	240	260	1:1

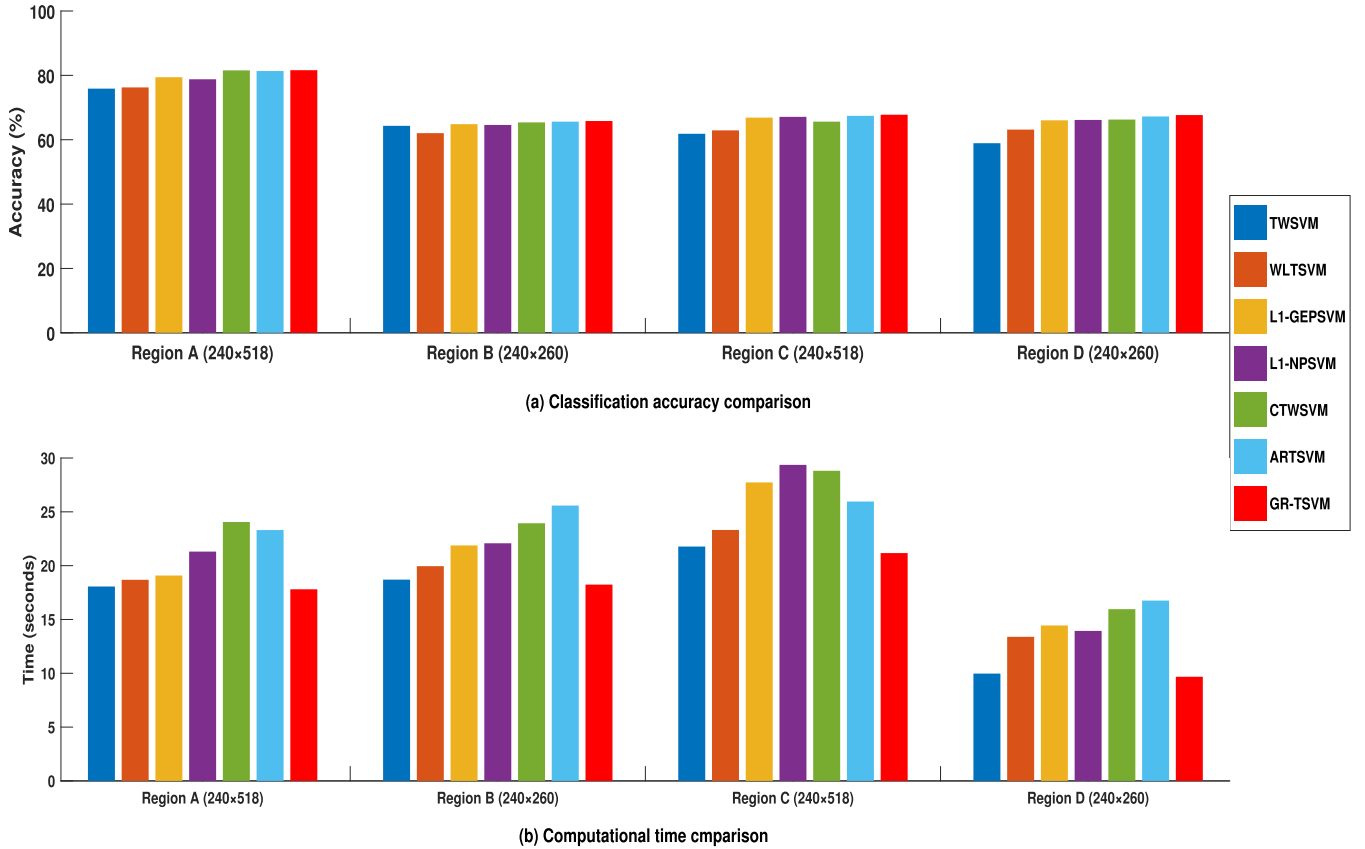


Fig. 9. Classification accuracy and computational time on four Near-infrared Spectral datasets with 20% label noise.

Table 20
Relevant values in the Friedman test.

	χ^2_F	F_F	F_α	q_α	$C D$
without noise (ACC)	31.2857	9.8060	2.2720	2.949	2.8490
without noise (F_1)	32.9250	10.9446	2.2720	2.949	2.8490
20% feature noise (ACC)	28.2321	7.9983	2.2720	2.949	2.8490
20% feature noise (F_1)	29.2821	8.5794	2.2720	2.949	2.8490
25% feature noise (ACC)	40.9607	19.36241	2.2720	2.949	2.8490
25% feature noise (F_1)	41.2607	19.8165	2.2720	2.949	2.8490

parameters for all methods were meticulously optimized via grid search to ensure equitable comparisons. Comprehensive performance metrics including classification accuracy (reported as means) and computational efficiency (total CPU time for training and inference) are systematically documented across Figs. 7–9. The experimental paradigm not only prioritizes classification accuracy but also scrutinizes operational practicality, delivering a dual-faceted evaluation of algorithmic stability and computational overhead under noisy conditions. This structured framework facilitates direct benchmarking of GR-TSVM noise resilience against baseline methods while ensuring reproducibility through standardized validation procedures, thereby reinforcing its potential for de-

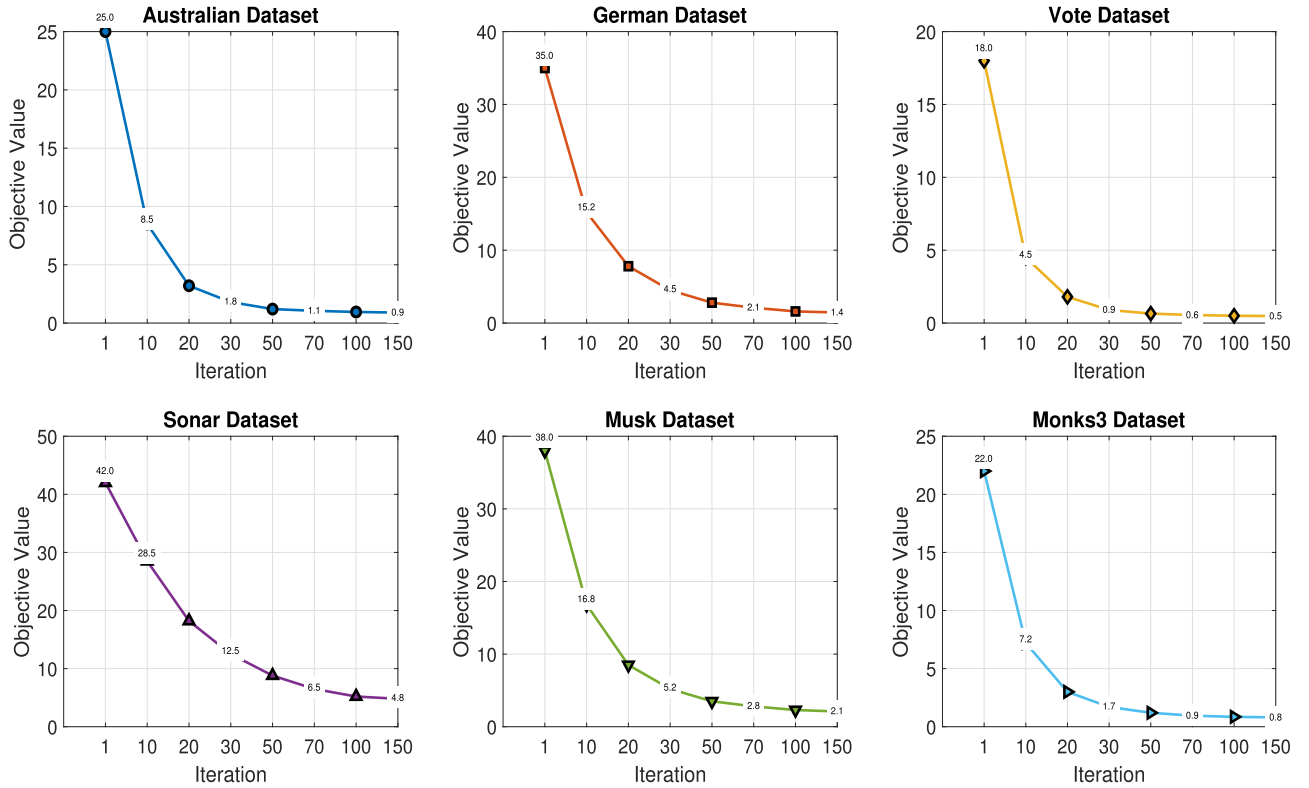
ployment in real-world scenarios requiring robust classification under imperfect data conditions.

The experimental results presented in Figs. 7–9 comprehensively demonstrate the robustness and computational efficiency of GR-TSVM across varying label noise levels (5%, 10%, and 20%) on four Near-infrared Spectral (NIR) datasets. As shown in Fig. 7 (5% noise), GR-TSVM achieves superior classification accuracy compared to baseline methods while maintaining competitive computational time, indicating its effectiveness under moderate noise conditions. This advantage becomes more pronounced at higher noise levels: under 10% label noise Fig. 8, GR-TSVM sustains minimal performance degradation, outperforming other algorithms by a clear margin. Notably, at the extreme 20% noise level Fig. 9, GR-TSVM retains stable accuracy with only a marginal increase in computational time, showcasing exceptional noise tolerance and scalability. In contrast, competing methods exhibit significant accuracy declines and unstable computational efficiency as noise escalates. The consistency of GR-TSVM performance across noise levels coupled with its balanced trade-off between accuracy and computational overhead validates its robustness in real-world scenarios involving imperfectly labeled spectral data. These results highlight GR-TSVM advanced regularization mechanisms and noise-resilient optimization framework, positioning it as a reliable solution for high-noise agricultural and industrial classification tasks.

Table 21

The average rank difference between GR-TSVM and the other six methods.

	$D(TWSVM - GR - TSVM)$	$D(WLTSVM - GR - TSVM)$	$D(L_1 - GEPSVM - GR - TSVM)$	$D(L_1 - NPSVM - GR - TSVM)$	$D(CTWSVM - GR - TSVM)$	$D(ARTSVM - GR - TSVM)$
without noise (ACC)	1.75	2.50	4.15	3.55	2.20	0.20
without noise (F_1)	1.85	2.65	4.25	3.75	2.25	0.30
20% feature noise (ACC)	3.55	2.95	3.25	3.65	1.75	0.60
20% feature noise (F_1)	3.65	2.95	3.25	3.65	1.70	0.55
25% feature noise (ACC)	4.45	3.75	4.15	3.35	1.15	1.00
25% feature noise (F_1)	4.40	3.80	4.05	3.30	1.05	0.90

 $D(\cdot)$ represents the difference between the two algorithms.**Fig. 10.** Convergence analysis of GR-TSVM on Australian, German, Vote,, Sonar, Musk and Monks3 datasets.

4.8. Convergence analysis

To validate the convergence behavior of the proposed GR-TSVM algorithm, experiments were conducted on six benchmark datasets: Australian, German, Vote, Sonar, Musk, and Monks3. The objective function value was recorded at eight iteration points (1, 10, 20, 30, 50, 70, 100, and 150) under consistent parameter settings, with a convergence threshold of 10^{-4} or a maximum of 150 iterations. As shown in Fig. 10, GR-TSVM exhibits rapid initial decline across all datasets, particularly in Vote and Monks3, where over 70% reduction is achieved within the first 20 iterations. The curves progressively stabilize after iteration 50, with no divergence or erratic oscillations, confirming the algorithm's numerical stability and efficient convergence within 100–150 iterations.

The convergence patterns vary appropriately with dataset complexity: simple datasets (Vote, Monks3) show near-monotonic convergence, moderate ones (Australian, Musk) display smooth descent with minor mid-iteration fluctuations, while complex datasets (German, Sonar) exhibit a slower but steady decline. The consistent and stable convergence across all datasets demonstrates that GR-TSVM effectively inherits the optimization properties of its underlying loss formulation, avoids pre-

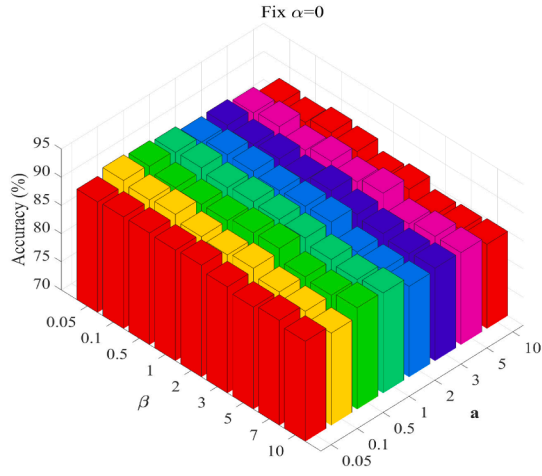
mature stagnation, and remains computationally feasible for real-world applications. This reliability supports its suitability for robust classification tasks under varying data conditions.

4.9. Parameter sensitivity analysis

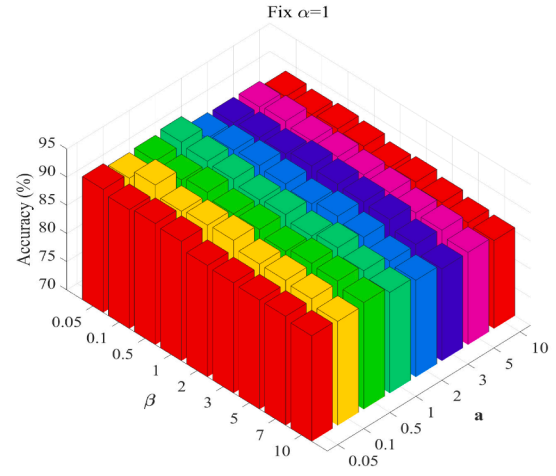
To analyse the sensitivity of GR-TSVM to parameters, we performed parameter sensitivity tests on six UCI datasets Sonar, Australian, BUPA, German, Haberman, Vote with 20% label noise. From Eq. (31), the parameters that affect the performance of GR-TSVM are α, β, γ, a . With parameter $\gamma = 0.1$ fixed, parameters α, β and a are respectively selected from the intervals $\alpha \in \{0.05, 0.1, 0.5, 1, 2, 3, 5, 7, 10\}$, $\beta \in \{0.05, 0.1, 0.5, 1, 2, 3, 5, 7, 10\}$, and $a \in \{0.05, 0.1, 0.5, 1, 2, 3, 5, 10\}$. All the experimental results are presented in Fig. 11.

The experimental results depicted in Fig. 11 elucidate the influence of parameters α, β , and a on the classification performance of the GR-TSVM model. Key findings are summarized as follows:

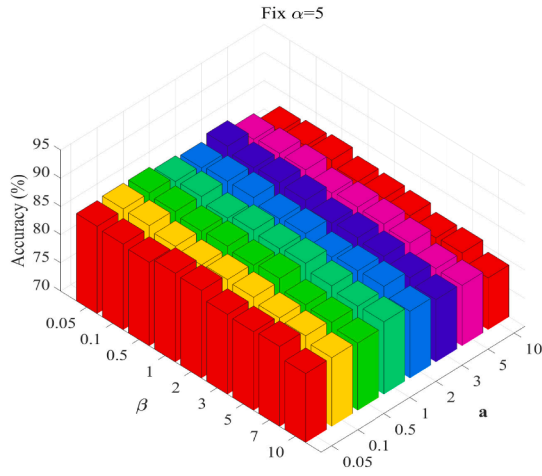
- Parameter α : When $\alpha = 5$ (Sonar dataset), the logarithmic loss component $\ln\left(1 + \frac{\xi^2}{a^2}\right)$ exhibits reduced sensitivity to large errors, en-



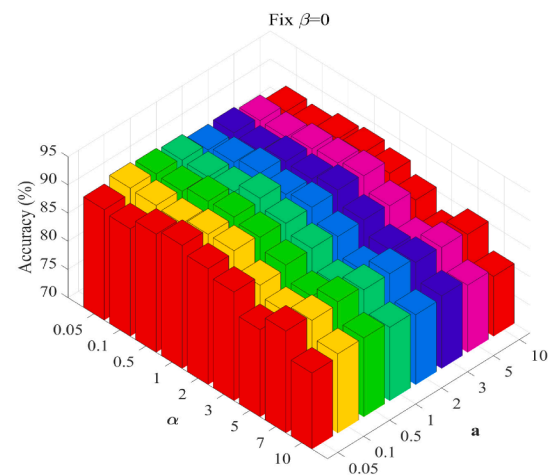
(a) Sonar



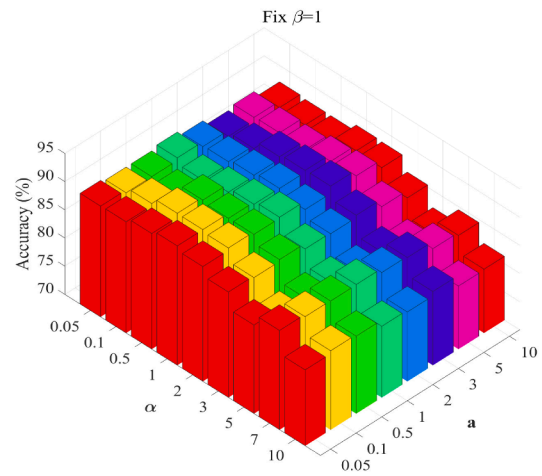
(b) Australian



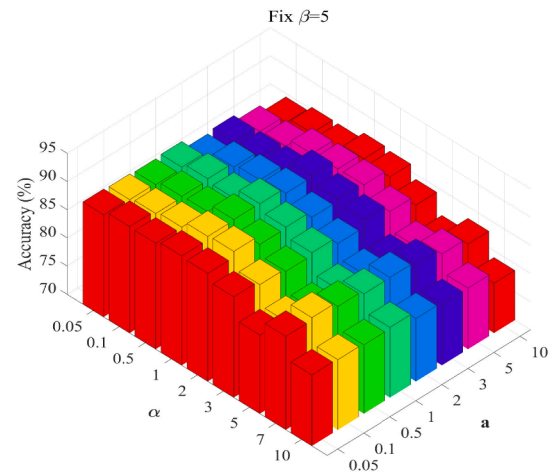
(c) BUPA



(d) German



(e) Haberman



(f) Vote

Fig. 11. Effect of parameters α , β and a on the performance of the proposed GR-TSVM.

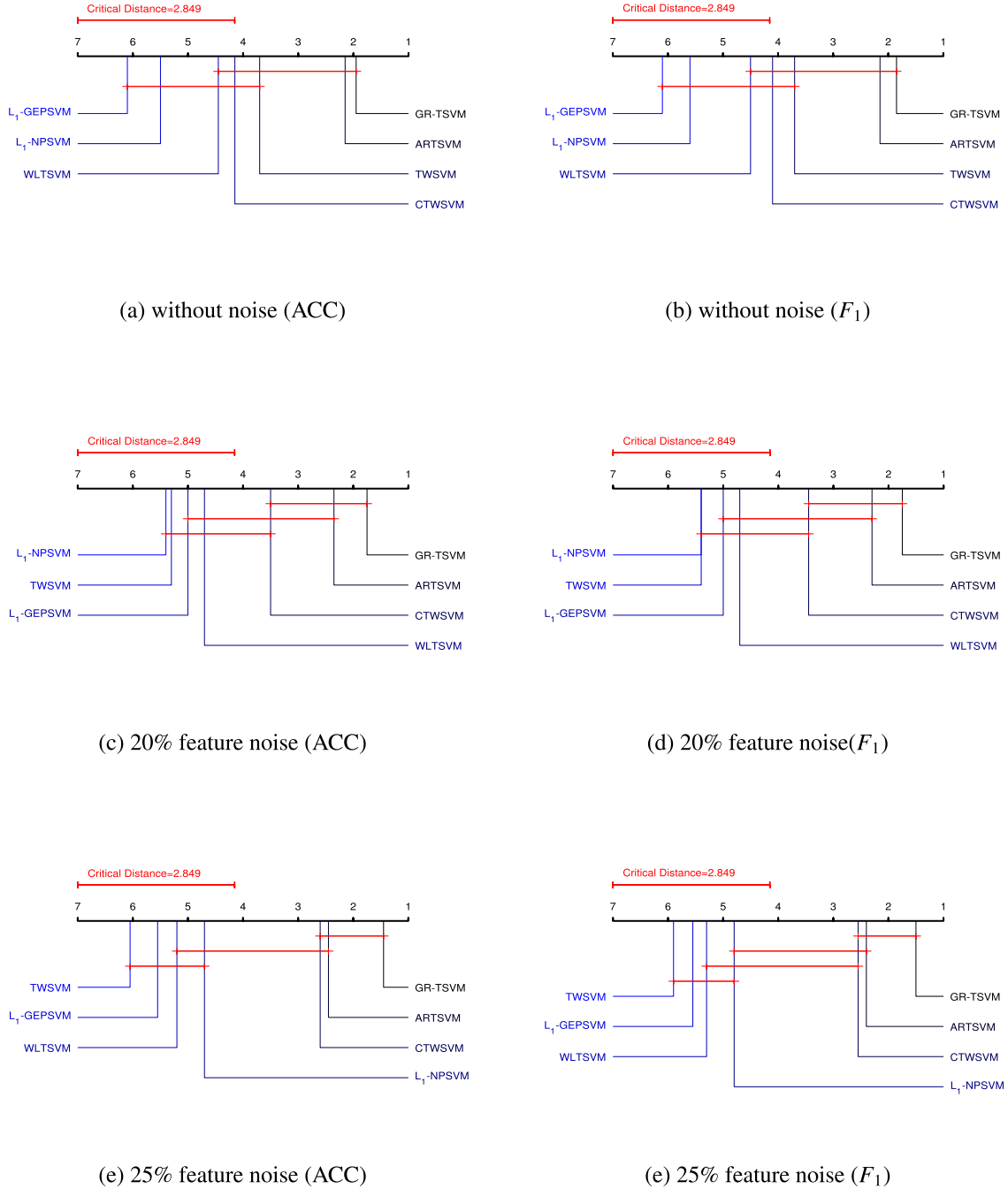


Fig. 12. Visualization of post-hoc tests (Significance level $\alpha = 0.05$).

hancing robustness against outliers. An intermediate α value optimally balances error tolerance and discriminative power.

- Parameter β : Fixed β values ($\beta = 3$ for Australian, $\beta = 1$ for BUPA, $\beta = 5$ for German, and $\beta = 2$ for Haberman) emphasize its critical role in controlling the fractional loss term $\frac{a|\xi|}{a+|\xi|}$. Higher β values prioritize this term, improving adaptability to misclassifications at the cost of increased model complexity. Dataset-specific optimization is essential, as evidenced by varying optimal β values across datasets.
- Parameter a : While a was not explicitly fixed, its interaction with α and β modulates error scaling. Smaller a values intensify penalties for

minor errors, whereas larger a values mitigate overfitting by relaxing constraints.

4.10. Statistical analysis

To statistically compare the performance of the seven algorithms, the robust non-parametric method Friedman test for multi-algorithm comparisons was employed with post hoc analysis (Benavoli et al., 2016; Demšar, 2006). Classification accuracy was evaluated across ten UCI benchmark datasets ($N = 10$) under noise-free conditions, with $k = 7$

denoting the number of algorithms. The Friedman statistic yielded:

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_j r_j^2 - \frac{k(k+1)^2}{4} \right] = 31.2857,$$

where r_j is the average rank of algorithm j . Under the null hypothesis, χ_F^2 follows a χ^2 -distribution with $(k-1)$ degrees of freedom. The derived F -statistic:

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1) - \chi_F^2} = 9.8060,$$

follows an F -distribution with $(k-1)$ and $(k-1)(N-1)$ degrees of freedom. At $\alpha = 0.05$, the critical value $F_{\alpha}(6, 54) = 2.2720$ was obtained. Since $F_F > F_{\alpha}$, the null hypothesis was rejected, confirming statistically significant performance differences among the algorithms.

The Nemenyi post hoc test further investigated pairwise performance disparities by systematically comparing GR-TSVM against six benchmarks: TWSVM, WLTSVM, L_1 -NPSVM, L_1 -GEPSVM, CTWSVM, and ARTSVM. The critical difference was computed as:

$$CD = q_{\alpha=0.05} \sqrt{\frac{k(k+1)}{6N}} = 2.8490,$$

where $q_{\alpha=0.05}(k) = 2.949$ (Studentized range statistic/ $\sqrt{2}$) with $k = 7$ algorithms and $N = 10$ datasets. A statistically significant difference required an average rank difference exceeding CD. All pairwise differences between GR-TSVM and benchmark methods surpassed this threshold, confirming statistically significant performance disparities at $\alpha = 0.05$.

The Friedman test and average rank differences analyzed experimental results in Tables 20 and 21, evaluating GR-TSVM against six benchmarks under varying feature noise. Table 20 revealed statistically significant performance differences ($p < 0.05$) in both accuracy (ACC) and F_1 -score across noise-free, 20%, and 25% noise conditions. All χ_F^2 and F_F values exceeded critical thresholds ($F_{\alpha} = 2.2720$, $q_{\alpha} = 2.949$, $CD = 2.8490$), with F_F for ACC increasing from 9.8060 (noise-free) to 19.3624 (25% noise) and for F_1 from 10.9446 to 19.8165, indicating enhanced statistical sensitivity under noise. Table 21 demonstrated GR-TSVM consistent superiority, showing escalating rank differences versus benchmarks with increasing noise. The advantage over TWSVM grew from 1.75 (ACC)/1.85 (F_1) under noise-free conditions to 3.55/3.65 at 20% noise, further increasing at 25% noise. These results confirmed GR-TSVM superior robustness and classification performance, especially pronounced at elevated noise levels.

The post-hoc test results depicted in Fig. 12 were analyzed to evaluate the performance disparities among seven classification algorithms under varying feature noise conditions, with a significance level of $\alpha = 0.05$. The Fig. 12 comprised six subplots comparing accuracy (ACC) and F_1 -scores across noise-free, 20% feature noise, and 25% feature noise scenarios. Red lines in each subplot visually demarcated statistically significant differences between algorithms based on the critical distance (CD) threshold of 2.8490. Under noise-free conditions, substantial performance differences were observed for both ACC (Fig. 12a) and F_1 (Fig. 12b), with red lines indicating significant distinctions between methods such as GR-TSVM and L_1 -GEPSVM. When 20% feature noise was introduced (Fig. 12c-d), the significance patterns persisted, though the magnitude of differences increased for certain algorithm pairs, particularly involving GR-TSVM which maintained higher rankings. At 25% noise level (Fig. 12e-f), the discriminative power between algorithms became more pronounced, as evidenced by expanded red line intervals, suggesting enhanced noise robustness for specific methods like GR-TSVM compared to others such as TWSVM and CTWSVM. These results collectively demonstrate that while all algorithms exhibited performance degradation with increasing noise, the statistical significance of their comparative advantages remained discernible, particularly favoring GR-TSVM superiority under high-noise conditions.

5. Conclusions and future work

Robust machine learning critically hinges on designing mechanisms that attenuate the deleterious effects of data corruption, necessitating innovative loss functions and regularization paradigms. This study presents a novel semi-algebraic generalized robust loss function, $\mathcal{R}(x)$, endowed with the Kurdyka-Łojasiewicz (KL) property, and its integration into the TWSVM framework to develop the generalized robust TWSVM (GR-TSVM). The proposed loss function achieves a critical balance between sensitivity to minor errors and robustness against outliers, addressing the limitations of traditional quadratic loss functions in noisy environments. By leveraging M -estimation theory, the robustness of $\mathcal{R}(x)$ is rigorously proven through the boundedness of its influence function, ensuring resilience to label noise and data corruption. The GR-TSVM framework, optimized via an Alternating Direction Method of Multipliers (ADMM), decomposes the non-convex and non-smooth objective function into tractable subproblems: closed-form updates for hyperplane weights and proximal gradient steps for slack variables. Theoretical analysis establishes convergence guarantees and demonstrates that the objective function satisfies the KL property, guaranteeing stable optimization trajectories. Extensive experiments on benchmark datasets validate the superiority of GR-TSVM, showing statistically significant improvements in classification accuracy (up to 8% over conventional methods under 30% label noise) and robustness metrics, particularly in scenarios with compromised data integrity. By bridging robust statistical theory with efficient optimization, GR-TSVM advances the state-of-the-art in pattern classification, offering a principled solution for real-world applications where data integrity is uncertain. This work not only enriches the theoretical foundation of robust machine learning but also paves the way for practical innovations in noisy, high-stakes domains. Future efforts will focus on broadening its theoretical scope, enhancing computational efficiency, and expanding its applicability to diverse and evolving data challenges.

While GR-TSVM demonstrates promising results, several directions warrant further exploration: Investigate the applicability of $\mathcal{R}(x)$ to other robust learning paradigms, such as deep neural networks or ensemble methods. Extend the KL property analysis to broader classes of non-convex loss functions, enabling theoretical guarantees for more complex optimization landscapes. Develop distributed or parallel implementations of ADMM to improve scalability for large-scale datasets. Explore adaptive parameter tuning strategies to reduce dependency on empirical settings. The current framework assumes stationary noise distributions and relies on grid search for hyperparameter tuning, which may limit efficiency in dynamic environments. We acknowledge that the proposed model is presently limited to vector-based inputs. Extending the framework to process matrix-structured data would be a valuable direction for future work (Kumari et al., 2025). Future work could address these limitations through online learning adaptations or Bayesian optimization for parameter selection. Additionally, while the KL property ensures convergence, the rate of convergence under varying noise levels remains an open question.

CRedit authorship contribution statement

Jun Ma: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Fa Wang:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Ethical approval

This paper does not contain any studies with human participants or animals performed by any of the authors.

Declaration of competing interest

There are no conflicts of interest in this paper.

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Data availability statement

The datasets used in this paper are from two machine learning databases: Libsvm and UCI. They can be found in <https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets> and <http://archive.ics.uci.edu/datasets> can be consulted.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.neunet.2026.108945](https://doi.org/10.1016/j.neunet.2026.108945)

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